

DTE NOTES CBE

Unit: Science and technology

1.0 PHILOSOPHY OF SCIENCE

1.1 NATURE OF SCIENCE

PROBLEM SOLVING PROCESS

Step 1

DEFINE THE SPECIFIC PROBLEM. -The starting point is recognizing and stating a very specific question/

Step 2

COLLECT INFORMATION BY MAKING OBSERVATIONS. -Gather background information on the problem, and make many observations, information obtained by using the senses.

Step 3

FORM A HYPOTHESIS. -Utilize your observations to form a hypothesis, a testable explanations for the problem.

Step 4

TEST THE HYPOTHESIS BY EXPERIMENTATION. -An experiment is a procedure that tests a hypothesis by the process of collecting information under controlled conditions. A controlled experiment involves two groups: the control group and the experimental group.

Step 5

ORGANIZE AND RECORD COLLECTED DATA. -The information obtained during the experimental procedure, the data, must be accurately recorded.

Step 6

DRAW CONCLUSIONS. -The data must be analyzed to determine if the results supported the hypothesis. If the hypothesis is not supported, propose a new hypothesis to test.

Example: The fire alarm is not working.

1) Observe/Define the problem: it does not beep when I push the button.

2) Hypothesis: it is caused by a dead battery.

3) Test: try a new battery.

4) Confirm/Disconfirm: the alarm now works. If it does not work, start over by testing another hypothesis like "it has a loose wire."

3.0 LIVING THINGS

3.1 Classification of living Things

Classification is putting organisms into groups.

Classification is based on the study of external characteristics of organisms.

It involves detailed observation of structure and functions of organisms.

Organisms with similar characteristics are put in one group.

Differences in structure are used to distinguish one group from another.

The magnifying lens is an instrument that assists in the observation of fine structure e.g. hairs by enlarging them. **Kingdoms**

Living organisms are classified into five kingdoms;

Monera,

Protocista,

Fungi,

Plantae

Animalia.

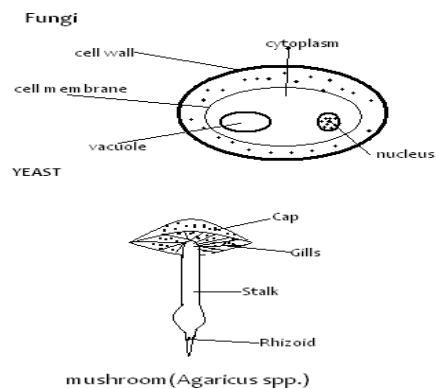
Kingdom Fungi

Some are unicellular while others are multicellular.

They have no chlorophyll.

Most are saprophytic e.g. yeasts, moulds and mushrooms.

A few are parasitic e.g. ***Puccinia graminea***.



Kingdom Monera (Prokaryota)

These are very small unicellular organisms.

They lack a nuclear membrane

do not have any bound membrane organelles.

Hence the name Prokaryota.

They are mainly bacteria, e.g. ***Vibrio cholerae***.

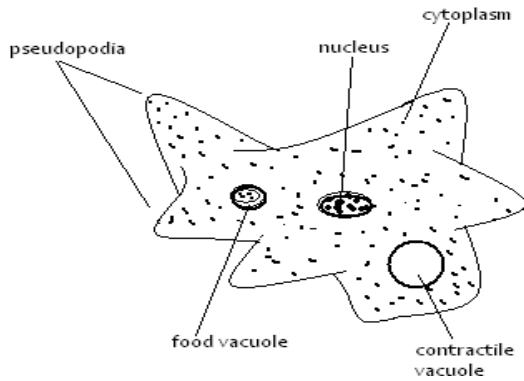
Kingdom Protocista

They are unicellular organisms.

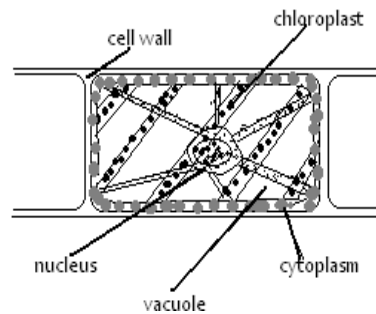
Their nucleus and organelles are surrounded by membranes (eukaryotic).

They include algae, slime moulds - fungi-like and protozoa

Amoeba-a protozoan(Protocist)



Spyrogyra-filamentous green algae



Kingdom Plantae

They are all multicellular.

They contain chlorophyll and are all autotrophic.

They include; Bryophyta (mossplant), Pteridophyta (ferns) and Spermatophyta (seedbearing plants).

Kingdom Animalia

These are all multicellular and heterotrophic.

Examples are annelida (earthworms),

mollusca (snails),

Example of Arthropods are ticks, butterflies.

Members of Chordata are fish, frogs and humans.

.

General Characteristics of Kingdoms

Organisms are classified into five kingdoms

Monera,

Protocista,

Fungi,

Plantae

Animalia.

Viruses do not fit neatly into any of the above kingdoms.

They are simple and not cellular.

They are metabolically inactive outside the host cell.

Most of them can be crystallised like chemical molecules.

Therefore they do not exhibit the characteristics of living organisms.

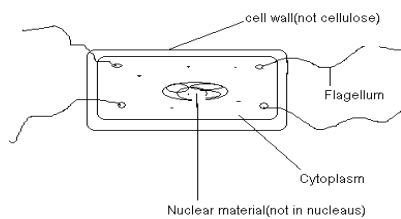
<i>Characteristic</i>	<i>Monera</i>	<i>Protocista</i>	<i>Fungi</i>	<i>Plantae</i>	<i>Animalia</i>
Cell type	Prokaryotic	Eucaryotic	Eucaryotic	Eucaryotic	Eucaryotic
Unicellular Multicellular	Unicellular	Unicellular and Multicellular	Unicellular and multicellular	Multicellular	Multicellular
Mode of Feeding	Autotrophic or heterotrophic by Absorption	Autotrophic or heterotrophic by absorption or Phagocytosis	Heterotrophs by absorption	Autotrophs	Heterotrophic by ingestion
Reproduction	Asexual by binary fission	Asexual binary fission, fragmentation, Sporulation	Asexual fission Fragmentation, sporulation	Asexual by Sporulation And Fragmentation Sexual	Sexual

Examples of Organisms in Each Kingdom and Their Economic Importance Kingdom Monera

General Characteristics

- Unicellular and microscopic
- Some single cells ,others colonial
- Nuclear material not enclosed within nuclear membrane-prokaryotic
- Have cell wall but not of cellulose.
- Have few organelles which are not membrane bound
- Mitochondria absent
- Mostly heterotrophic, feeding saprotrophically or parasitically,some are autotrophic.
- Reproduction mostly asexual through binary fission
- Most of them are anaerobes but others are aerobes
- Most move by flagella

A generalised drawing of a bacterium



Examples include ***Escherichia coli***, ***Vibrio cholerae*** and ***Clostridium tetani***.

Spherical known as Cocci.

Rod shaped - e.g. *Clostridium tetani*

Spiral shaped e.g. spirilla

Coma shaped- Vibrios -e.g., *Vibrio cholerae*.

Kingdom Protocista

Examples include ;

Algae such as spirogyra, Chlamydomonas, euglena, Sargassum

And protozoa such as amoeba, paramecium and Trypanosoma.

General Characteristics

They are said to be eukaryotic since their nucleus is bound by a membrane

Most are mobile, and use flagella, cilia and pseudopodia.

Some are sessile.

They reproduce mainly asexually, by binary fission, fragmentation and sporulation.

Some reproduce sexually by conjugation.

Some are heterotrophic e.g. paramecium.

Others are autotrophic e.g. spirogyra.

Spirogyra: They have spiral chloroplast.

They are green, thread-like filaments

Chlamydomonas:

This is a unicellular green algae and has a cup shaped chloroplast.

They move towards light using the flagella

Cilia assist the organism to move.

The shape is due to the presence of a thin flexible pellicle.

Kingdom Fungi

Multicellular fungi are made of thread-like structures called hyphae (singular hyphae) that form a mycelium.

.e.g. *Saccharomyces cerevisiae* (bread yeast).

Others include *Penicillium*, *Rhizopus*, and edible mushroom

Kingdom Plantae General

Characteristics

They are multicellular and eukaryotic.

They are photosynthetic and have a pigment chlorophyll.

Their cells have cellulose cell walls.

They reproduce sexually, others asexually.

Kingdom Plantae has three major divisions:

Bryophyta,

Pteridophyta

Spermatophyta.

Division Bryophyta

These include mosses and liverworts.

Plant body is not differentiated into root, stem and leaves.

They have simple structures which resemble leaves and stems.

They have rhizoids for absorbing water and anchoring the plant to substratum.

Life cycle consists of two morphologically different plants, the gametophyte and sporophyte. They show alternation of generations.

The gamete producing gametophyte is the persistent plant.

The sporophyte is attached to the gametophyte and is nutritionally dependent on it.

They lack vascular system.

Sexual reproduction is dependent on water.

Division Pteridophyta:

These include ferns and horsetails.

General Characteristics

They have root and shoot system.

Leaves are compound known as fronds, they have a vascular system.

They show alternation of generations whereby the spore bearing sporophyte is the main plant.

Spores are borne in clusters on the underside of leaves making sori.

The gametophyte is an independent minute structure called prothallus which is short-lived.

Sexual reproduction is dependent on water. ***Division Spermatophyta***

These are the seed bearing plants.

General Characteristics

Plant body is differentiated into root, stem and leaves.

Vascular tissue consists of xylem and phloem.

Sexual reproduction is independent of water.

Male gametophyte (pollen grain) germinates and grows to reach female gametophyte.

They are divided into two sub-divisions:

Gymnosperms

Angiosperms.

Gymnosperms

These are cone-bearing plants.

Naked seeds.

They are trees and shrubs.

Xylem consists of tracheids only.

Examples; pine, cypress and spruce.

They show xerophytic characteristics like having needle-like leaves.

Angiosperms

Seeds are enclosed within a fruit.

They comprise trees, shrubs and herbs.

Xylem consists of vessels and tracheids.

These are the most advanced plants.

Angiosperms has two classes;

Monocotyledonae

Dicotyledonae.

Comparison of Dicotyledonae and Monocotyledonae

<i>Dicotyledonae</i>	<i>Monocotyledonae</i>
Embryo has two cotyledons.	Embryo has one cotyledon.
Leaves are broad and have network of veins.	Leaves are long with parallel veins (have leaf sheath)
T.S. of root has no pith.	T.S. of root has pith.
Have tap root system.	Have fibrous root system.
Cross section of stem reveals vascular bundles arranged in a ring.	Cross section of stem reveals vascular bundles scattered all over.
Vascular cambium present and have secondary growth.	Vascular cambium absent and do not have secondary growth.
Flower parts in four, five or multiples of these.	Flower parts in three or multiples of three.
Examples: herbs e.g. tomatoe; shrubs e.g. tea,	Examples: grass, wheat, sugar-cane.

Economic Importance of Spermatophyta

They are a source of food for humans and other animals.

Source of fuel- wood fuel and charcoal.

Source of timber for building and for paper.

Ornamental plants.

Useful in textile industry.

Kingdom Animalia

Most animals move from place to place in search of food. Major phyla are:

Platyhelminthes (Tapeworm).

Nematoda (Ascaris).

Annelida (Earthworm).

Mollusca (Snails).

Arthropoda

chordata

Phylum Arthropoda

Distinguishing Characteristics

They have jointed appendages, which are specialised for various functions.

Their body is covered by a hardened exoskeleton made of chitin.

It is shed at intervals to allow for growth.

They have jointed body parts.

Most are divided into head, thorax and abdomen.

Some have two body parts,

General Characteristics Body is segmented.

They have bilateral symmetry.

Gaseous exchange is through tracheal system, book lungs or gills which opens to the outside through spiracles.

Aquatic forms use gills.

Reproduction is mainly sexual.

They have an open circulatory system.

Phylum Arthropoda divided into five classes;

Crustacea,

Arachnida,

Chilopoda,

Diplopoda

Insecta

This division is based on:

The number of limbs.

Presence and number of antennae.

Number of body parts.

Class Crustacea

Most of them are aquatic, a few are terrestrial found in moist places e.g., woodlouse.

Distinguishing Characteristics

Two body parts head and thorax are fused to form cephalothorax and an abdomen .

They have two pairs of antennae; one is small and branched, the other is long.

They have five or more pairs of limbs.

Some of these are modified for other functions e.g., locomotion, feeding and defence.

Exoskeleton hardened with deposits of calcium carbonate i.e. carapace.

Other Characteristics

Mouthparts include a pair of mandibles and two pairs of maxillae.

Gaseous exchange is through gills.

They have a pair of compound eyes.

Most crustaceans are free-living but a few are parasitic e.g., barnacles.

Examples are cray-fish and crab.

Class Arachnida

Members are carnivorous and paralyse prey using poison produced from poisonclaws.

Distinguishing Characteristics

The body has two parts: cephalothorax and abdomen.

Cephalothorax is head fused to thorax.

A pair of chelicerae, on ventral side of cephalothorax.

They have four pairs of walking legs.

They have no antennae.

Instead they have a pair of short pedipalps which are sensitive to touch.

Most arachnids use book lungs for gaseous exchange.

Other characteristics include simple eyes.

Examples include garden spider, ticks, scorpions.

Class Chilopoda e.g. Centipede

Distinguishing Characteristics

The body has 2 body parts, a head and trunk.

The body is elongate, and has 15 or more segments.

Has a pair of legs on each segment.

The body is dorso-ventrally flattened.

Other characteristics include:

Head has a pair of antennae.

Gaseous exchange through tracheal system. Are carnivorous.

Class Diplopoda e.g. Millipede

Distinguishing Characteristics

Has two parts: head, short thorax and a trunk .

Body elongate with 9-100 segments.

Has two pairs of legs on each segment.

They have a cylindrical body.

Gaseous exchange is by tracheal system.

Other characteristics:

Head has a pair of antennae.

Are herbivorous.

Class Insecta

Distinguishing Characteristics

Body is divided into three body parts head, thorax and abdomen.

They have three pairs of legs ..

Most insects have a pair or two of wings.

Other characteristics include:

A pair of antennae.

They breathe through spiracles, and gaseous exchange is through tracheal system.

Phylum Chordata

This name is derived from the term notochord.

This is a long flexible rod-like structure.

The more familiar chordates are known as vertebrates.

In vertebrates the notochord exists only in embryonic stages of development which in later stages is replaced by a vertebral column.

Main Characteristics of Vertebrates

Members of the phylum have a notochord in early stages of development.

They have visceral clefts - which are slits perforating the body wall at the pharynx.

In fish these slits become gills while in higher chordates these slits are only present in embryo.

They have a dorsal, hollow nerve cord.

It develops into a brain at the anterior and spinal cord at the posterior end.

The spinal cord is enclosed within the vertebral column.

They have segmented muscle blocks known as myotomes on either side of the body.

They possess a post-anal tail although rudimentary in some.

They have a closed circulatory system.

The heart is ventrally located.

They possess an internal skeleton.

The main classes of phylum chordata are;

Fish,
Amphibians,
Reptiles,
Birds Mammal

Fish

These are the fishes.

Some fish have a skeleton made of cartilage e.g. the shark.

`Others like Tilapia have a bony skeleton.

Distinguishing Characteristics

They are aquatic.

Movement is by means of fins.

They have a streamlined body.

They have a lateral line for sensitivity.

Their heart has two chambers, the auricle and ventricle - simple circulatory system.

Other Characteristics

Their body temperature changes according to the temperature of the environment.

They are ectothermic (poikilothermic).

Body covered with scales.

They have gills for gaseous exchange.

Exhibit external fertilization.

Amphibians

Larval forms are aquatic while adults are terrestrial.

Adults return to water for breeding e.g. frogs, toads, newts, salamanders.

Distinguishing Characteristics

Skin is soft and without scales.

They have four well developed limbs.

The hind limbs are longer and more muscular than forelimbs.

The limb can be used for walking, jumping and swimming

Gaseous exchange is through the skin, gills and lungs.

Middle ear is present.

Other Characteristics

They have a three-chambered heart with two atria and one ventricle.

Fertilisation is external.

They are ectothermic (poikilotherms).

Reptiles

Examples are snakes, crocodiles, lizards, chameleons, tortoises and turtles.

Distinguishing Characteristics

The skin is dry and is covered by horny scales.

Fertilisation is internal.

Some species eggs contain a lot of yolk and have either leathery or calcareous shells.

They have a double circulatory system.

The heart has three chambers - two atria and a partly divided ventricle.

However crocodiles have a four chamber heart.

Other Characteristics

They are ectothermic (poikilothermic).

Have 2 pairs of limbs.

They use lungs for gaseous exchange.

Birds

These are birds.

They are terrestrial and arboreal and others are aquatic
e.g. flamingo, goose, ostrich, penguin, hawk, dove.

Distinguishing Characteristics

Body is covered by feathers and legs with horny scales.

They have two pairs of limbs.

Fore limbs modified to form wings for flight.

Hind limbs are for walking or swimming.

The mouth is a protruding beak.

They have hollow bones.

They have double circulation with a four-chambered heart (2 atria, 2 ventricles).

They have lungs for gaseous exchange.

Lungs are connected to air sacs in bones.

Fertilisation is internal.

They lay eggs with calcareous brittle shell.

They have constant body temperatures hence are homoiotherms (endothermic).

Mammals

They are arboreal e.g. tree-squirrels,

Others terrestrial e.g. humans

Others are aquatic e.g. dolphins and whales.

Distinguishing Characteristics

They have mammary glands hence name of the class.

Body is covered with fur or hair.

Their teeth are differentiated into four types (heterodont dentition).

They have external ear-pinna.

Most have sweat glands.

They have a diaphragm that separates the body cavity into thoracic and abdominal.

Other Characteristics

Internal fertilisation - most give birth.

They have a double circulatory system with a four-chambered heart.

They are endothermic (homoiotherms) .

Eg Duck-billed Platypus (egg-laying mammal)

Eg.Kangaroo (pouched mammal)

The young are born immature and are nourished in a pouch with milk from mammaryglands.

Placental Mammals

They give birth to fully developed young ones which are fed on milk from mammaryglands.

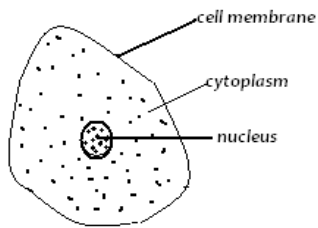
Some are aquatic. e.g. dolphins, whale,

Others are flying e.g, bat;

Most are terrestrial e.g. rabbits, elephants, buffalo, giraffe, antelope, cow, humanbeing.

3.2 THE CELL

The cell is the basic unit of an organism.



A typical animal cell

All living organisms are made up of cells.

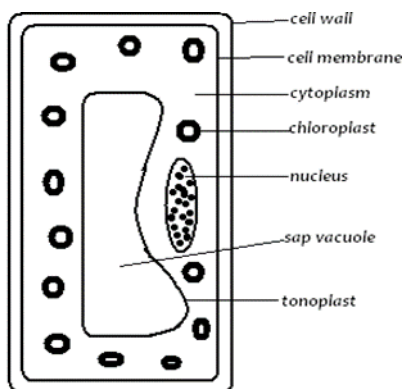
Some organisms are made up of one cell and others are said to be multicellular.

Other organisms are made of many cells and are said to be multicellular.

Cells are too little to see with the naked eye.

They can only be seen with the aid of a microscope.

Cell Structure as Seen Through the Light Microscope



Generalised plant cell

The cell as seen above has the following:

Cell membrane (Plasma membrane):

This is a thin membrane enclosing cell contents.

It controls the movement of substances into and out of the cell.

Cytoplasm:

This is a jelly-like substance in which chemical processes are carried out.

Scattered all over the cytoplasm are small structures called organelles.

Like an animal cell, the plant cell has a cell membrane, cytoplasm and a nucleus.

vacuole.

Plant cells have permanent, central **vacuole**. It contains cell sap where sugars and salts are stored.

Cell wall:

This is the outermost boundary of a plant cell.

It is made of cellulose.

Between the cells is a middle lamella made of calcium pectate.

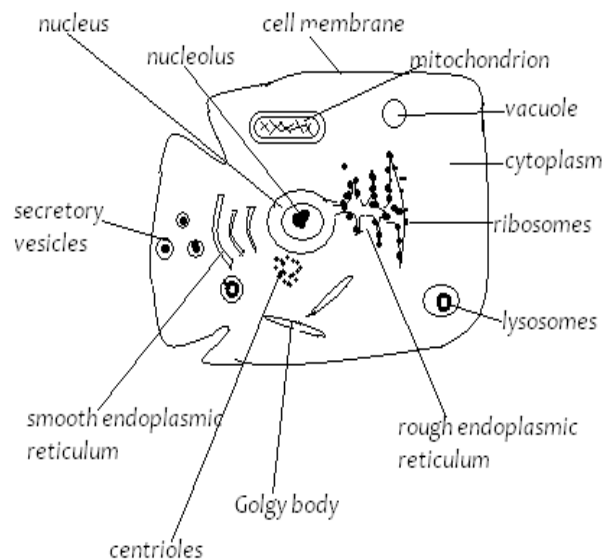
Chloroplasts;

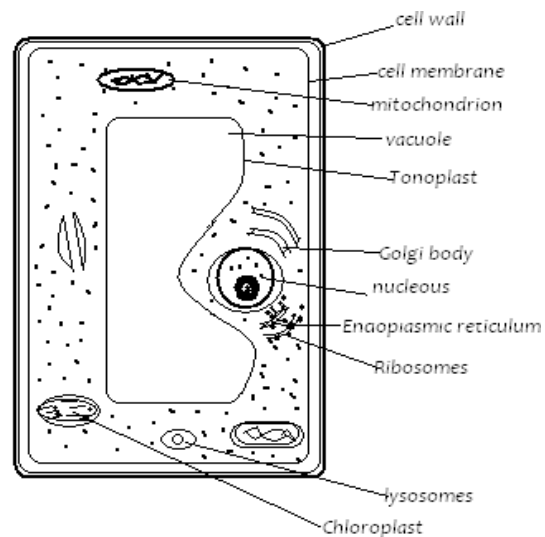
With special staining techniques it is possible to observe **chloroplasts**.

These are structures which contain chlorophyll, the green pigment responsible for trapping light for photosynthesis

Cell Structure as Seen Through Electron Microscope

A generalised animal cell as seen under Electron microscope





A generalised plant cell as seen under the electron microscope

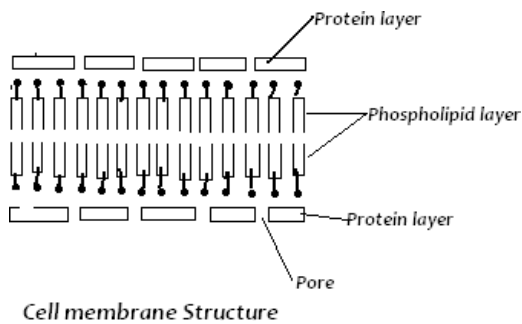
The Plasma Membrane

Under the electron microscope, the plasma membrane is seen as a double layer.

This consists of a lipid layer sandwiched between two protein layers.

This arrangement is known as the unit membrane and shows two lipid layers with proteins within.

Substances are transported across the membrane by active transport and diffusion.



The Endoplasmic Reticulum (ER)

This is a network of tubular structures extending throughout the cytoplasm of the cell.

It serves as a network of pathways through which materials are transported from one part of the cell to the other.

An ER encrusted with ribosomes it is referred to as **rough endoplasmic reticulum**.

An ER that lacks ribosomes is referred to as **smooth endoplasmic reticulum**.

The rough endoplasmic reticulum transports **proteins** while the smooth endoplasmic reticulum transports **lipids**.

The Ribosomes

These are small spherical structures attached to the ER.

They consist of protein and ribonucleic acid (RNA).

They act as sites for the synthesis of proteins.

Golgi Bodies

Golgi bodies are thin, plate-like sacs arranged in stacks and distributed randomly in the cytoplasm.

Their function is packaging and transportation of glycol-proteins.

They also produce lysosomes.

Mitochondria

Each mitochondrion is a rod-shaped organelle.

Made up of a smooth outer membrane and a folded inner membrane.

The foldings of the inner membrane are called **cristae**.

They increase the surface area for respiration.

The inner compartments called the matrix.

Mitochondria are the sites of cellular respiration, where energy is produced.

Lysosomes

These are vesicles containing hydrolytic enzymes.

They are involved in the breakdown of micro-organisms, foreign macromolecules and damaged or worn-out cells and organelles ..

The Nucleus

The nucleus is surrounded by a nuclear membrane which is a unit membrane.

The nuclear membrane has pores through which materials can move to the surrounding cytoplasm.

The nucleus contains proteins and nucleic acid deoxyribonucleic acid (DNA) and RNA.

The chromosomes are found in the nucleus.

They are the carriers of the genetic information of the cell.

The nucleolus is also located in the nucleus but it is only visible during the non-dividing phase of the cell.

The Chloroplasts

These are found only in photosynthetic cells.

Each chloroplast consists of an outer unit. membrane enclosing a series of interconnected membranes called lamellae.

At various points along their length the lamellae form stacks of disc like structures called grana.

The lamellae are embedded in a granular material called the stroma.

The chloroplasts are sites of photosynthesis.

The light reaction takes place in the lamellae while the dark reactions take place in the stroma.

Comparison between animal cell and plant cell

<i>Plant Cell</i>	<i>Animal Cell</i>
• Has a cell wall and a cell membrane. • Nucleus at periphery. • Have • Has a large central vacuole. • Are usually large. • Are regular in shape. • Has no centriole. • Stores starch, oils and protein.	• Has cell membrane only. • Nucleus at the center. • Have no chloroplasts. • Has no vacuoles, they are small and scattered. • Are usually small. • Irregular in shape. • Has centrioles. • Store glycogen and fats.

Cell Specialisation

Cells are specialised to perform different functions in both plants and animals.

Example;

Palisade cells have many chloroplasts for photosynthesis.

Root hair cells are long and thin to absorb water from the soil.

Red blood cells have haemoglobin which transports oxygen.

Sperm cells have a tail to swim to the egg.

Multicellular organisms cells that perform the same function are grouped together to form a tissue.

Each tissue is therefore made up of cells that are specialised to carry out a particular function.

Animal Tissues-Examples of animal tissues

Type of Tissue	Functions	Characteristics
1. Epithelial Tissue	Covering. allowing movement of materials	
(a) Squamous	Covering of internal organs. lining for	Thin flat cells.
(b) Columnar	Secretion. absorption e.g. in the	Cells that are longer than they are wide.
(c) stratified	Covering surfaces, protection e.g. the	Several layers of epithelial cells.
(b Smooth)	cover internal organs; allow	Are spindle-shaped.
(d) Cuboidal	Absorption e.g. in the kidney	squamous. cuboidal or cube like cells.
(c) Cardiac	peristalsis. Cause contraction of the heart.	controlled by involuntary contract rhythmically; are
2. Muscular Tissue	Contraction, bringing about	Contists of units called (ability to contract is within)
3. Supporting Tissue	Support the body. provide a rigid framework, protect soft tissue.	Cells that produce hard
(a) Cartilage		
(b Bone)		
4. Blood	Transport of materials. Protection	A complex tissue consisting of of cells suspended in a fluid
5. Nerve Tissue	Receive stimuli and transmit impulses; co-ordinate body activities	Consists of cells called which are interconnected axons to enable transmission

Plant Tissues

Example of plant tissues

<i>Type of Tissue</i>	<i>Functions</i>	<i>Characteristics</i>
1. <i>Meristematic</i>	Undergo division and cause e.g. increase in length and girth	Small thin-walled cells, contain a lot of cytoplasm; found mostly at the tip of shoots and roots.
2. <i>Parenchyma</i>	Photosynthesis gaseous support; storage.	Thin walled cells; vary in shape and size; many intercellular spaces.
3. <i>Collenchyma</i>	Strengthening.	Thickened walls; no intercellular spaces; found in cortex of stems.
4. <i>Sclerenchyma</i>	Strengthening.	Vary in shape; thick cell walls; are usually dead.
5. <i>Vascular</i> (a) Xylem (b) Phloem	Transport materials. Transport of water and mineral Transport of organic materials (manufactured food).	Tubular vessels and tracheids joined end to end. Sieve elements joined to each other through sieve pores.

Organs

An organ is made up of different tissues

e.g. the heart, lungs, kidneys and the brain in animals and roots, stems and leaves in plants.

Organ systems

Organs which work together form an organ system.

Digestive, excretory, nervous and circulatory in animals and transport and support system in plants.

organism

Different organ systems form an organism.

CELL PHYSIOLOGY

Meaning of cell physiology

The term physiology refers to the functions that occur in living organisms.

Cell physiology refers to the process through which substances move across the cell membrane.

Several physiological processes take place inside the cell.e.g. respiration.

Oxygen and glucose required enter the cell while carbon (IV) oxide and water produced leave the cell through the cell membrane.

Structure and properties of cell membrane

The cell membrane is the protective barrier that shelter cellular contents.

Movement of all substances into and out of the cells takes place across the cell membrane.

It is made up of protein and lipid molecules.

Lipid molecules have phosphate group attached to it on one end.

They are then referred to phospholipids.

The phospholipids are arranged to form a double layer.

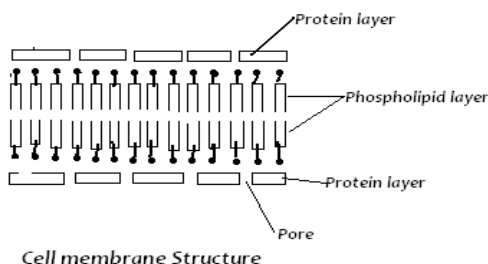
The ends with phosphate group face outwards.

the proteins are scattered throughout the lipid double layer.

Some of these proteins act as carrier molecules that channel some material in and outside the cells.

The cell membrane allows certain molecules to pass through freely while others move through with difficulty and still others do not pass through at all.

This is selective permeability and the cell membrane is described as semi-permeable.



Properties of cell membrane

Permeability

The cell membrane is semi-permeable.

it allows small molecules that are soluble in lipid to pass through with more ease than water soluble molecules.

this is due to the presence of the phospholipids double layer.

Polarity

The cell membrane has electrical charges across its surface. It has positive charged ions on the outside and negatively charged ions on the inside. This property contributes to electrical impulses sent along nerve cells.

Sensitivity to changes in temperature and pH

Very high temperatures destroy the semi-permeability nature of the cell membrane because the proteins are denatured by extreme pH values have the same effect on the membrane permeability.

Physiological processes

Some of the physiological processes include diffusion, osmosis and active transport.

Diffusion

Diffusion is the movement of molecules or ions from a region of high concentration to a region of low concentration aided by a concentration gradient..

Diffusion continues to occur as long as there is a difference in concentration between two regions (concentration gradient).

Stops when an equilibrium is reached i.e., when the concentration of molecules is the same in both regions.

Diffusion is a process that occurs inside living organisms as well as the external environment..

Does not require energy.

Factors Affecting Diffusion

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Concentration Gradient

An increase in the concentration of molecules at one region results in a steeper concentration gradient which in turn increases the rate of diffusion.

Temperature

High temperature increases kinetic energy of molecules. They move faster hence resulting in an increase in rate of diffusion, and vice versa.

Size of Molecules or Ions

The smaller the size of molecules or ions, the faster their movement hence higher rate of diffusion.

Density

The denser the molecules or ions diffusing, the slower the rate of diffusion, and vice versa.

Medium

The medium through which diffusion occurs also affects diffusion of molecules or ions. For example, diffusion of molecules through gas and liquid media is faster than through a solid medium.

Distance

This refers to the thickness or thinness of surface across which diffusion occurs. Rate of diffusion is faster when the distance is small i.e., thin surface.

Surface Area to Volume Ratio

The larger the surface area to volume ratio, the faster the rate of diffusion.

For example, in small organisms such as Amoeba the surface area to volume ratio, is greater hence faster diffusion than in larger organisms.

Role of Diffusion in Living Organisms

Some processes that depend on diffusion include the following:

Gaseous exchange: Movement of gases through respiratory surfaces is by diffusion.

Absorption of materials into cells Cells obtain raw materials and nutrients from the surrounding tissue fluid and blood through diffusion, e.g., glucose needed for respiration diffuses from blood and tissue fluid into cells.

Excretion: Removal of metabolic waste products like carbon (IV) oxide, and ammonia out of cells is by diffusion.

Absorption of the end-products of digestion from the intestines is by diffusion.

Osmosis

Osmosis is the movement of water molecules from a region of high water concentration to a region of low water concentration through a semi-permeable membrane.

Osmosis is a special type of diffusion that involves the **movement of water molecules only** and **not solute molecules**.

Osmosis takes place in cells across the cell membrane as well as across non-living membranes e.g. cellophane or visking tubing which are also semi-permeable, It is purely a physical process.

?

Factors Affecting Osmosis

Size of solute molecules-

Osmosis occurs only when solute molecules are too large to pass through a semi-permeable membrane.

Concentration Gradient .

Osmosis occurs when two solutions of unequal solute concentration are separated by a semi-permeable membrane.

Temperature ,.

High temperatures increase movement of water molecules hence influence osmosis. However, too high temperatures denature proteins in cell membrane and osmosis stops.

Pressure

Increase in pressure affects movement of water molecules. As pressure increases inside a plant cell, osmosis decreases.

Roles of Osmosis in Living Organisms

The following processes depend on osmosis in living organisms:

Movement of water into cells from the surrounding tissue fluid and also from cell to cell.

Absorption of water from the soil and into the roots of plants.

Support in plants especially herbaceous ones, is provided by turgor pressure, which results from intake of water by osmosis.

Absorption of water from the alimentary canal in mammals.

Re-absorption of water in the kidney tubules.

Opening and closing stomata.

Water Relations in Plant and Animal Cells

The medium (solution) surrounding cells or organisms is described by the terms **hypotonic**, **hypertonic** and **isotonic**.

A solution whose solute concentration is more than that of the cell sap is said to be **hypertonic**.

A cell placed in such a solution loses water to the surroundings by osmosis.

A solution whose solute concentration is less than that of the cell sap is said to be **hypotonic**.

A cell placed in such a solution gains water from the surroundings by osmosis.

A solution which has the same solute concentration as the cell sap is said to be **isotonic**. When a cell is placed in such a solution there will be no net movement of water either into or out of the cell.

Osmotic Pressure

The term osmotic pressure describes the tendency of the solution with a high solute concentration to draw water into itself when it is separated from distilled water or dilute solution by a semi-permeable membrane.

Osmotic pressure is measured by an osmometer.

When plant cells are placed in distilled water or in a hypotonic solution, the osmotic pressure in the cells is higher than the osmotic pressure of the medium.

This causes the water to enter the cells by osmosis.

The water collects in the vacuole which increases in size.

As a result the cytoplasm is pushed outwards and it in turn presses the cell membrane next to the cell wall.

This builds up water pressure (hydrostatic pressure) inside the cell.

When the cell is stretched to the maximum, the cell wall prevents further entry of water into the cell.

Then the cell is said to be **fully** turgid.

The hydrostatic pressure developed is known as **turgor** pressure.

Plasmolysis

When a plant cell is placed in a hypertonic medium, it loses water by osmosis.

The osmotic pressure of the cell is lower than that of the medium.

The vacuole decreases in size and the cytoplasm shrinks as a result of which the cell membrane loses contact with the cell wall.

The cell becomes flaccid. The whole process is described as plasmolysis.

Incipient plasmolysis is when a cell membrane just begins to lose contact with the cell wall.

Plasmolysis can be reversed by placing the cell in distilled water or hypotonic solution. However, full plasmolysis may not be reversed if cell stays in that state for long.

Wilting

The term wilting describes the drooping of leaves and stems of herbaceous plants after considerable amounts of water have been lost through transpiration.

It is observed in hot dry afternoons or in dry weather.

This is when the amount of water lost through transpiration exceeds the amount absorbed through the roots.

Individual cells lose turgor and become plasmolysed and the leaves and stems droop.

The condition is corrected at night when absorption of water by the roots continues while transpiration is absent.

Eventually, wilting plants may die if the soil water is not increased through rainfall or watering.

Water Relations in Plants and Animals

Haemolysis

Haemolysis is the bursting of cell membrane of red blood cells releasing their haemoglobin.

It occurs when red blood cells are placed in distilled water or hypotonic solution.

- This is because the cell membrane does not resist further entry of water by osmosis after maximum water intake.

Crenation

Takes place when red blood cells are placed in hypertonic solution.

They lose water by osmosis, shrink and their shape gets distorted.

Animal cells have mechanisms that regulate their salt water balance (osmoregulation) to prevent above processes that lead to death of cells.

An Amoeba placed in distilled water, i.e. hypotonic solution, removes excess water using a contractile vacuole.

The rate of formation of contractile vacuoles increases.

Active Transport

Active transport is the movement of solutes such as glucose, amino acids and mineral ions;

From an area of their low concentration to an area of high concentration.

It is movement against a concentration gradient and therefore energy is required.

As such it only takes place in living organisms.

The energy needed comes from respiration.

Certain proteins in the cell surface membrane responsible for this movement are referred to as **carrier proteins or channel proteins**.

The shape of each type of carrier protein is specific to the type of substances conveyed through it.

It has been shown that the substance fits into a particular slot on the protein molecule,

As the protein changes from one form of shape to another the substance is moved across and energy is expended.

Factors Affecting Active Transport ***Availability of oxygen***

Energy needed for active transport is provided through respiration.

An increase in the amount of oxygen results in a higher rate of respiration.

If a cell is deprived of oxygen active transport stops .

Temperature

Optimum temperature is required for respiration, hence for active transport.

Very high temperatures denature respiratory enzymes.

Very low temperatures inactivate enzymes too and active transport stops.

Availability of carbohydrates

Carbohydrates are the main substrates for respiration.

Increase in amount of carbohydrate results in more energy production during respiration and hence more active transport.

Lack of carbohydrates causes active transport to stop.

Metabolic poisons

Metabolic poisons e.g. cyanide inhibit respiration and stops active transport due to lack of energy.

Role of Active Transport in Living Organisms

Processes requiring active transport:

Absorption of mineral salts from the soil into plant roots.

Absorption of end products of digestion e.g. glucose and amino acids from the digestive tract into blood stream.

Excretion of metabolic products e.g. urea from the cells.

Re-absorption of useful substances and mineral salts back into blood capillaries from the kidney tubules.

Sodium-pump mechanism in nerve cells.

Re-absorption of useful materials from tissue fluid into the blood stream.

Practical Activities

Experiment to Demonstrate Diffusion

Various coloured substances such as: dyes, plant extracts and chemicals like potassium manganate are used.

Potassium manganate (VII) crystals are introduced to the bottom of a beaker filled with water using a glass tubing or drinking straw which is then removed.

Observations are made and the disappearance of the crystals and subsequent uniform colouring of water noted.

Experiment to Show Osmosis using Living Tissue

Irish potato tubers are peeled and scooped out to make hollow space at the centre.

Sucrose solution is placed inside the hollow, and the potato tuber placed in a beaker or petri-dish

with distilled water. A control is set using a boiled potato.
Another one using distilled water inside hollow in place of sugar solution.
The experiment is left for 3 hours to 24 hours.

Experiment to Demonstrate Turgor and Plasmolysis in Onion Epidermal Cells

Two strips of onion epidermis are obtained.

One is placed on a slide with distilled water while the other is placed on a slide with 25% sucrose solution and a coverslip placed on top of each.

The mounted epidermis is observed under low power microscope and then left for 30 minutes.

After 30 minutes, observations are made again.

The cells in distilled water have greatly enlarged. Cells in 25% sucrose have shrunk.

3.3 THE HUMAN BODY SYSTEM



Human Body Organ Systems

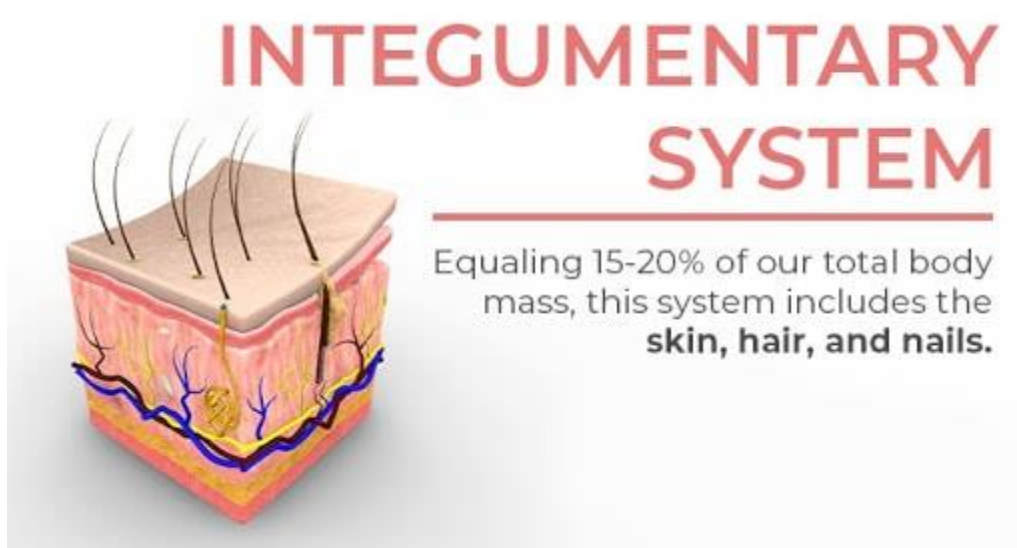
Body systems are groups of organs and tissues that work together to perform important jobs for the body. Some organs may be part of more than one body system if they serve more than one function. Other organs and tissues serve a purpose in only one body system. The 11 organ systems include; **the integumentary system, skeletal system, muscular system, lymphatic system, respiratory system, digestive system, nervous system, endocrine system, cardiovascular system, urinary system, and reproductive systems.**

All body systems are necessary for a complex [organism](#) to be able to survive and reproduce. **This article will focus on the systems of the human body;** similar systems are required by all animals, but the details of how they accomplish their tasks may vary.

Functions that must be performed by an animal to stay alive include:

- Absorbing oxygen for use in [cellular respiration](#)
- Excreting carbon dioxide produced during cellular respiration
- Ingesting and processing food to obtain sugars and other nutrients.
- Transporting necessary substances, such as oxygen and nutrients, to all cells in the body
- Clearing toxic waste products from the body.
- Responding to changes in the environmental conditions
- Protecting the organs from the environment.

- Fighting pathogens
- Additionally, for a [species](#) to survive, its individuals must be able to reproduce.



Integumentary System (Skin, Hair, Nails)

The integumentary system is the largest organ of the body, equaling 15-20% of our total body mass.

It acts as a barrier to physical, chemical, and biological agents.

The skin prevents water loss and regulates body temperature. It transmits the senses of touch, pain, and pleasure and maintains body temperature by secreting sweat.

The hair lubricates the scalp, which secretes pheromones and cools or warms our heads.

The nails protect our fingers, which are a major tool used for protecting ourselves and providing ourselves with food, shelter, and sensations.

The skin leaves us most vulnerable when it is compromised by open wounds, allowing infectious agents into the body.



Musculoskeletal System

Skeletal System (Bones, Joints)

The skeletal system supports and protects the body's internal organs.

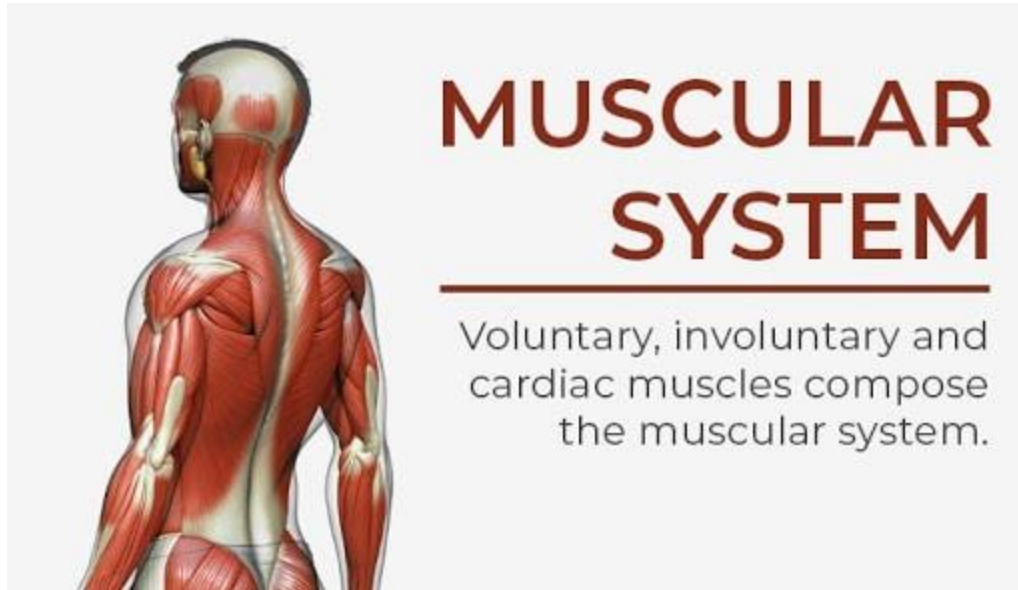
The ribs protect the abdominal organs, which are both vulnerable to injury and dangerous to our well being when injured.

The skull protects our brain which controls all functions of our bodies and minds.

The skeleton provides the framework and shape to our bodies. It also connects to our major muscles to allow movement.

Bones store minerals such as calcium and create blood cells in the soft bone tissue called marrow.

Bones can break easily without enough calcium and are subject to such diseases as arthritis; cancers; [scoliosis](#); osteoporosis, gout; bursitis; fractures and breaks; and amputations.



Muscular system (Cardiac, Smooth, and Skeletal Muscles)

Cardiac muscles are found in the heart and power the actions that maintain blood flow through our body;

Smooth, or involuntary muscles are found in the heart and organs, they surround the internal organs and are responsible for their movement such as moving food through the digestive tract; and

Skeletal, or voluntary muscles, are responsible for carrying out the actions and movements caused by messages sent from our brains through our nervous system. Skeletal muscles are also responsible for maintaining posture and producing heat. When muscles lack appropriate levels of oxygen they can cramp and tear, creating pain. When not used they can atrophy and become useless. Diseases and disorders of the muscular system include muscular dystrophy; fibromyalgia; tendinitis; multiple sclerosis; and muscle strain or sprains; hernias.

LYMPHATIC SYSTEM

Red bone marrow, thymus, lymphatic vessels, thoracic duct, spleen, and lymph nodes make up this system responsible for fluid transport.



Lymphatic System (Red Bone Marrow, Thymus, Lymphatic Vessels, Thoracic Duct, Spleen, Lymph Nodes)

This system transports clean fluids in our body back to the blood and drains excess fluids and debris from the tissues and cells of the body.

It also houses the white blood cells (lymphocytes) involved in protecting our bodies from infection.

Diseases and disorders specific to the lymphatic system include [anemias](#); leukemia; tuberculosis of the lymph nodes; Hodgkin's disease; and other blood disorders.

Lymphatic System/Immune System

The lymphatic system includes the circulatory and immune systems. **The lymphatic system is responsible for absorbing fat from the digestive system, maintaining fluid balance, and supporting the cells of the immune system.**

Every living thing needs to be able to fight invasion by foreign particles. This is because every organism that is made of delicious carbohydrates, proteins, and lipids has another organism that wants to take advantage. Some of these organisms are large predators, but most are microscopic pathogens that can easily fit inside us.

In the case of animals, we have white blood cells that can specifically target and destroy invading pathogens. These white blood cells are made in our bone marrow and stored in our blood and our lymphatic systems.

The lymphatic system is a circulatory system separate from the cardiovascular system. It carries water, white blood cells, and other substances, but it does not have red blood cells or platelets. Because it is

not the body's main carrier of oxygen, [lymph](#) can move more slowly than the bloodstream – giving the white blood cells more time to find and attack invaders.

“Lymph nodes” are nexuses in the lymphatic system where white blood cells can cluster and attack invading pathogens. Sometimes, when we're sick, our lymph nodes – such as those behind the ears, under the jaw, and in the armpits and groin – can become painful and swollen as our immune systems fight the infections in these nodes.

Immune Responses

Immune response is the production of antibodies in response to antigens.

An antigen is any foreign material or organism that is introduced into the body and causes the production of antibodies.

Antigens are protein in nature.

An antibody is a protein whose structure is complementary to the antigen.

This means that a specific antibody deals with a specific antigen to make it harmless.

When harmful organisms or proteins invade the body, lymphocytes produce complementary antibodies, while bone marrow and thymus gland produce more phagocytes and lymphocytes respectively.

Types of Immunity

There are two types of **immunity; natural and artificial.**

Natural Immunity is also called innate immunity.

It is inherited from parent to offspring.

Artificial Immunity can be natural or induced.

When attacked by diseases like chicken pox, measles and mumps, those who recover from these diseases develop resistance to any subsequent infections of the same diseases.

This is natural acquired immunity.

Artificial Acquired Immunity:

When attenuated (weakened) or dead microorganisms are introduced into a healthy person.

The lymphocytes synthesize the antibodies which are released into the lymph and eventually reach the blood.

The antibodies destroy the invading organisms.

The body retains 'memory' of the structure of antigen.

Rapid response is ensured in subsequent infections.

Vaccines generally contain attenuated disease causing organisms.

Artificial Passive Acquired Immunity:

Serum containing antibodies is obtained from another organism, and confers immunity for a short duration.

Such immunity is said to be passive because the body is not activated to produce the antibodies.

Importance of Vaccination

A vaccine is made of attenuated, dead or nonvirulent micro-organism that stimulate cells in the immune system to recognise and attack disease causing agent through production of antibodies. Vaccination protects individuals from infections of many diseases like smallpox, tuberculosis and poliomyelitis.

Diseases like smallpox, tuberculosis and tetanus were killer diseases but this is no longer the case. Diphtheria Pertussis Tetanus (DPT) vaccine protects children against diphtheria, whooping cough and tetanus.

Bacille Calmette Guerin (BCG) vaccine is injected at birth to children to protect them against tuberculosis.

Measles used to be a killer disease but today, a vaccine injected into children at the age of nine months prevents it.

At birth children are given an inoculation through the mouth of the poliomyelitis vaccine.



Respiratory System – (Nasal Cavity, Pharynx, Larynx, Trachea, Bronchus, Lung)

This system maintains our breathing.

It supplies the body with oxygen for cellular respiration by collecting oxygen in the lungs and disposes of carbon dioxide by breathing out the waste product. It also provides our functions of speech and smell.

Diseases and disorders of the respiratory system include allergies; rhinitis and sinusitis; laryngitis; COPD; pleurisy; bronchitis; emphysema; asthma; sarcoidosis; fibrosis; asbestosis; pulmonary vascular diseases;

fungal or bacterial infections of the lungs; [sleep apnea](#); tuberculosis of the respiratory system; and lung, throat, and other respiratory cancers.

DIGESTIVE SYSTEM

For absorption of nutrients, the digestive system includes the oral cavity, esophagus, liver, stomach, small intestine, large intestine, rectum and anus.

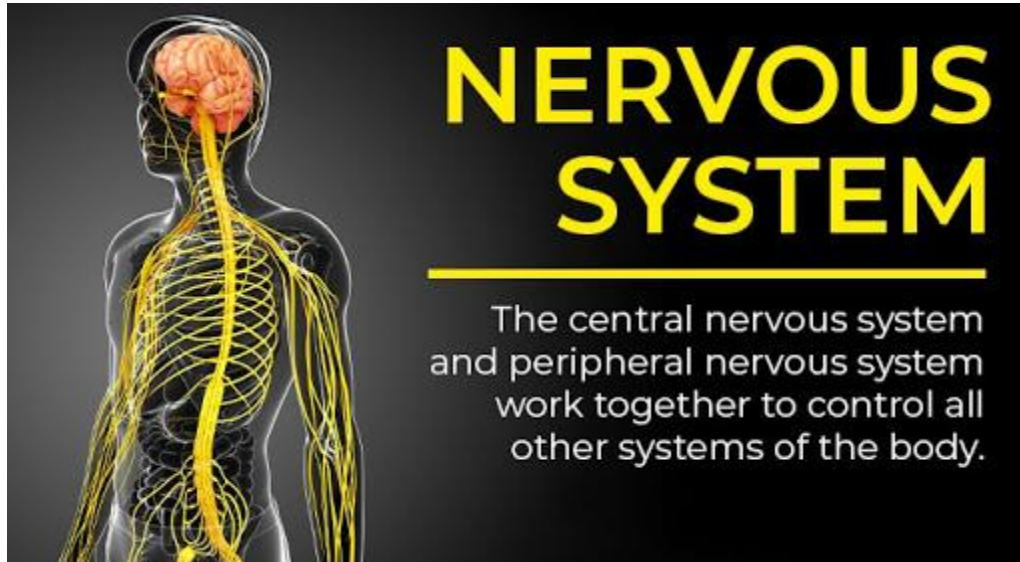


Digestive system (Oral Cavity, Esophagus, Liver, Stomach, Small Intestine, Large Intestine, Rectum, Anus)

Beginning with our mouths, this system is responsible for the breaking down and absorption of nutrients and the elimination of the waste not utilized by the body.

It is responsible for identifying which minerals, vitamins, and other essentials from the foods we eat can be absorbed and utilized or stored by the body and which are to be disposed of, and carrying out those functions.

Diseases and disorders of the digestive system include diverticulitis; gastritis; pancreatitis; cholecystitis; cirrhosis; hepatitis; liver cancer; irritable bowel or colon syndromes; Crohn's disease; and [hemorrhoids](#).



Nervous System– (Brain, Spinal Cord, Nerves)

This system is actually made up of two distinct parts; the central nervous system (CNS) and the peripheral nervous system.

The central nervous system is made up of the brain and spinal cord, and the peripheral nervous system is made up of all the nerves that lead into and out of the CNS to other parts of the body.

The entire nervous system controls all of the other systems of the body, such as digestion and cardiac rhythm, and responds to internal and external changes, such as activating muscles and breathing.

It also transmits information to the brain, such as pain and external sensations.

Diseases and disorders of the nervous system include paralysis; Parkinson's Disease; palsy, embolisms; thrombosis; arteriosclerosis; polio; myelitis; ALS; meningitis; Multiple Sclerosis; muscular tics; Huntington's Disease; cancers of the brain, spinal cord or nerves; epilepsy, seizure disorders; narcolepsy; migraine headaches; peripheral neuropathy; and traumatic brain injury.

ENDOCRINE SYSTEM

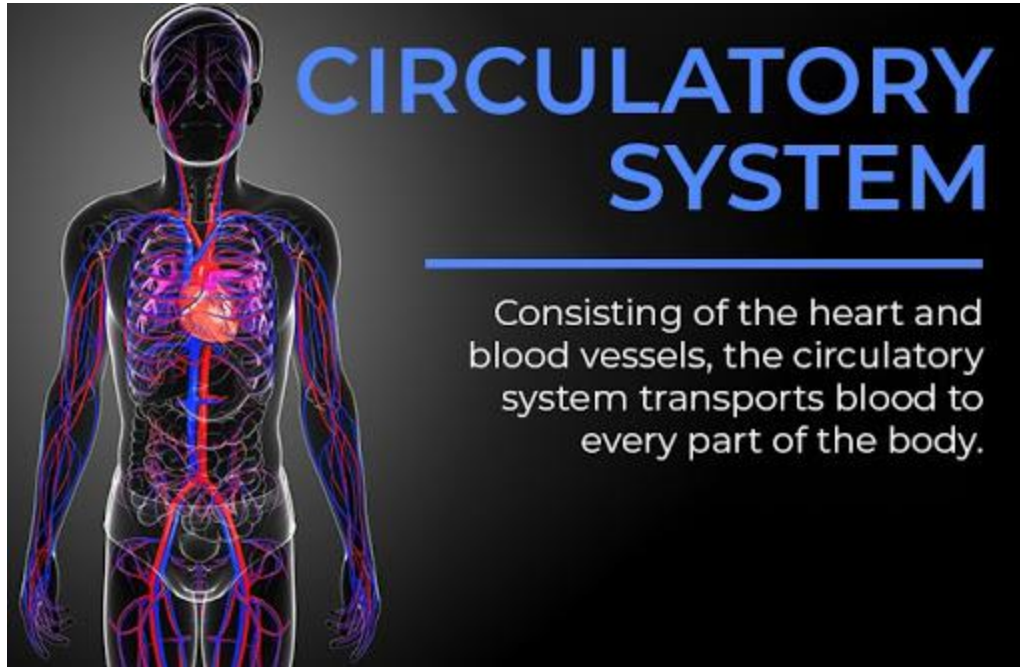
Responsible for releasing hormones, this system includes the pineal gland, pituitary gland, thyroid, thymus, adrenal gland, pancreas, ovary and testis.



Endocrine System– (Pineal Gland, Pituitary Gland, Thyroid Gland, Thymus, Adrenal Gland, Pancreas, Ovary, Testis)

The glands of the endocrine system secrete chemicals called hormones that regulate most of the processes in our bodies such as growth, reproduction, metabolism, and even the control of the amount of glucose in our blood.

Diseases and disorders of the endocrine system include [Type 1 & Type 2 Diabetes](#), hypoglycemia, Addison's disease; Cushing's syndrome; hyper/hypothyroidism, thyroid cancer; and other active cancers of the endocrine glands.



Cardiovascular (Circulatory) System– (Heart, Blood Vessels)

The heart, made of cardiac muscle, pumps blood and blood vessels such as arteries and veins, transport the blood to every part of our body providing organs and muscles with nourishment.

The blood carries oxygen, carbon dioxide, nutrients, waste and more throughout the body.

Diseases and disorders associated with the cardiovascular system include myocardial infarction; coronary bypass surgery; arrhythmias; valve replacements; pacemakers; transplants; [heart diseases](#) (ischemic; hypertensive; arteriosclerotic); hypertension; aneurysms; fistulas; arteriosclerosis; anaphylaxis shock; varicose veins; cold weather injuries; and sarcomas of the blood vessels.

URINARY SYSTEM

Responsible for eliminating waste, this system includes the kidneys, ureter, urinary bladder, and urethra.



Urinary system (Kidney, Ureter, Urinary Bladder, Urethra)

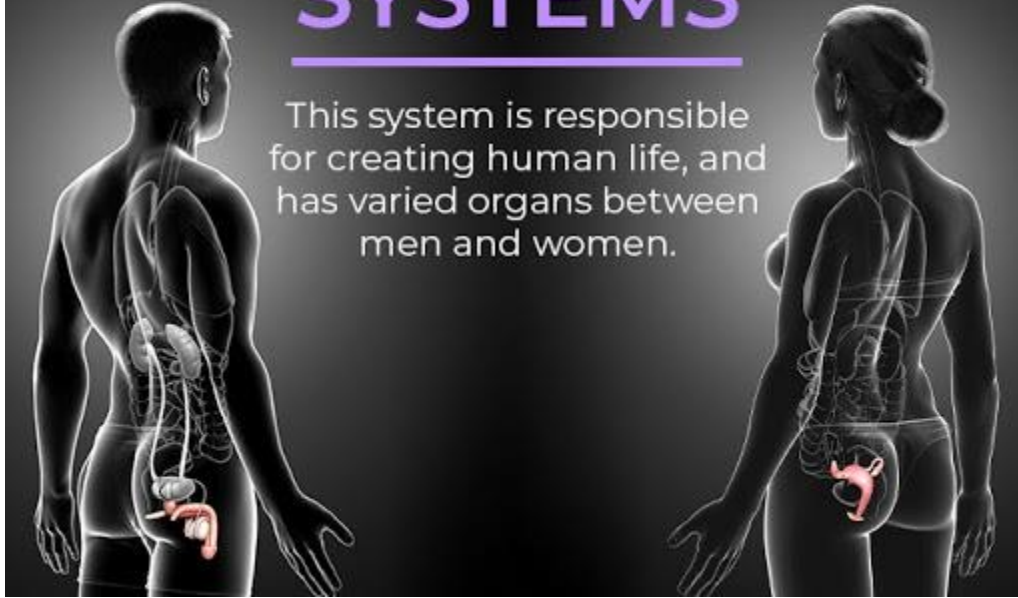
This system is responsible for eliminating waste products of metabolism and other materials from the body that are of no use.

The system is also responsible for maintaining the balanced fluid volume in our bodies by regulating the amount of water that is excreted, maintaining the concentrations of electrolytes, and normal pH levels of the blood.

Diseases and disorders of the urinary system include nephrosis, [bladder cancer](#), urethritis, bedwetting (enuresis), urinary and kidney stones and infections, renal failure, incontinence, blood in the urine, and interstitial cystitis.

REPRODUCTIVE SYSTEMS

This system is responsible for creating human life, and has varied organs between men and women.



Reproductive Systems

MALE (prostate gland, penis, testis, scrotum, ductus deferens)

FEMALE (Mammary glands, ovary, uterus, vagina, fallopian tube)

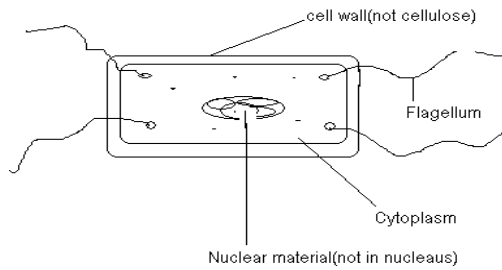
The reproductive system mainly functions to create human life.

Ovaries produce female sex hormones and eggs. Eggs are fertilized in the fallopian tube by sperm then travel to the uterus, which provides the site for growth. The mammary glands produce milk for the newborn.

[Diseases and disorders associated with the female reproductive systems](#) include breast cancer, removal of breast or lumpectomy; hysterectomy; pregnancy complications; endometriosis; ovarian or cervical cancers; and diseases or injuries of the vulva, vagina, cervix, uterus, fallopian tubes, or ovaries.

3.4 MICROORGANISMS

A generalised drawing of a bacterium



Examples include *Escherichia coli*, *Vibrio cholerae* and *Clostridium tetani*.

Spherical known as Cocci.

Rod shaped - e.g. *Clostridium tetani*

Spiral shaped e.g. spirilla

Coma shaped- Vibrios -e.g., *Vibrio cholerae*.

Economic importance of bacteria Benefits to man include:

They are used in food processing e.g., *Lactobacillus* used in processing of cheese, yoghurt.

Involved in synthesis of vitamin Band K, in humans and breakdown of cellulose in herbivores.

Genetic Engineering

Bacteria are easily cultured and are being used for making antibiotics, aminoacids andenzymes e.g. amylase, and invertase e.g., *Escherichia coli*.

Nutrient cycling:

Saprophytes

They are involved in decomposition of dead organic matter.

They are useful in the nitrogen cycle.

Nitrogen fixing and nitrifying bacteria.

They increase soil fertility.

Modern sewage works use bacteria in treatment of sewage.

Cleaning oil spills in oceans and lakes.

Harmful Effects

Bacteria cause disease:

To humans (e.g. Cholera).

To animals (e.g. Anthrax).

Bacteria cause food spoilage.

Others cause food poisoning e.g. *Salmonella*.

Denitrifying bacteria reduce soil fertility e.g., *Pseudomonas denitrificans*.

Protoctista

Examples include ;

Algae such as spirogyra, Chlamydomonas, euglena, Sargassum

And protozoa such as amoeba, paramecium and Trypanosoma.

General Characteristics

They are said to be eukaryotic since their nucleus is bound by a membrane

Most are mobile, and use flagella, cilia and pseudopodia.

Some are sessile.

They reproduce mainly asexually, by binary fission, fragmentation and sporulation.

Some reproduce sexually by conjugation.

Some are heterotrophic e.g. paramecium.

Others are autotrophic e.g. spirogyra.

Economic importance of protoctista

Algae are the primary producers in aquatic food chains.

They release a lot of oxygen to the atmosphere.

Some cause human diseases like malaria and amoebic dysentery ,sleeping sickness

Some are source of food for humans e.g. sargassum is a source of iodine

Skeletons of diatoms used in paint making.

Spirogyra: They have spiral chloroplast.

They are green, thread-like filaments

Chlamydomonas:

This is a unicellular green algae and has a cup shaped chloroplast.

They move towards light using the flagella

Cilia assist the organism to move.

The shape is due to the presence of a thin flexible pellicle.

Fungi

Multicellular fungi are made of thread-like structures called hyphae (singular hyphae) that form a mycelium.

.e.g. Saccharomyces cerevisiae (bread yeast).

Others include Penicillium, Rhizopus, and edible mushroom

Economic Importance of Fungi Beneficial

Effects

Some fungi are used as food e.g. mushrooms.

Some are decomposers which enhance decay to improve soil fertility - recycling of nutrients e.g., toadstools.

Some are useful in brewing and bread making e.g., yeast. Yeast is used as food - a rich source of Vitamin B.

Some are useful in production of antibiotics e.g., *Penicillium griseofulvin*.

Used in sewage treatment e.g., *Fusarium spp*.

Harmful Effects

Some cause food poisoning by producing toxic compounds e.g. *Aspergillus flavus* which produces aflatoxins.

Some cause food spoilage, fabric and wood spoilage through decomposition.

Some cause diseases to humans e.g., athlete's foot and ringworms.

Others cause diseases to plants e.g., potato blight (Irish potatoes) rust in tomatoes and smuts in cereals.

3.5 GASEOUS EXCHANGE IN PLANTS AND ANIMALS

Necessity for Gaseous Exchange in Living Organisms

Living organisms require energy to perform cellular activities.

The energy comes from breakdown of food in respiration.

Carbon (IV) oxide is a by product of respiration and its accumulation in cells is harmful which has to be removed.

Most organisms use oxygen for respiration which is obtained from the environment.

Photosynthetic cells of green plants use carbon (IV) oxide as a raw material for photosynthesis and produce oxygen as a byproduct.

The movement of these gases between the cells of organisms and the environment comprises gaseous exchange.

The process of moving oxygen into the body and carbon (IV) oxide out of the body is called breathing or ventilation.

Gaseous exchange involves the passage of oxygen and carbon (IV) oxide through a respiratory surface.

Diffusion is the main process involved in gaseous exchange.

Gaseous Exchange in Plants

Oxygen is required by plants for the production of energy for cellular activities.

Carbon (IV) oxide is required as a raw material for the synthesis of complex organic substances.

Oxygen and carbon (IV) oxide are obtained from the atmosphere in the case of terrestrial plants and from the surrounding water in the case of aquatic plants.

Gaseous exchange takes place mainly through the stomata.

Structure of Guard Cells

The stoma (stomata - plural) is surrounded by a pair of guard cells.

The structure of the guard cells is such that changes in turgor inside the cell cause changes in their shape.

They are joined at the ends and the cell walls facing the pore (inner walls) are thicker and less elastic than the cell walls farther from the pore (outer wall).

Guard cells control the opening and closing of stomata.

Mechanism of Opening and Closing of Stomata

In general stomata open during daytime (in light) and close during the night (darkness).

Stomata open when osmotic pressure in guard cells becomes higher than that in surrounding cells due to increase in solute concentration inside guard cells. Water is then drawn into guard cells by osmosis. Guard cells become turgid and extend.

The thinner outer walls extend more than the thicker walls.

This causes a bulge and stoma opens.

Stomata close when the solute concentration inside guard cells become lower than that of surrounding epidermal cells.

The water moves out by osmosis, and the guard cells shrink i.e. lose their turgidity and stoma closes.

Proposed causes of turgor changes in guard cells.

Accumulation of sugar.

Guard cells have chloroplasts while other epidermal cells do not.

Photosynthesis takes place during daytime and sugar produced raises the solute concentration of guard cells.

Water is drawn into guard cells by osmosis from surrounding cells.

Guard cells become turgid and stoma opens.

At night no photosynthesis occurs hence no sugar is produced.

The solute concentration of guard cells falls and water moves out of the guard cells by osmosis.

Guard cells lose turgidity and the stoma closes.

pH changes in guard cells occur due to photosynthesis.

In day time carbon (IV) oxide is used for photosynthesis. This reduces acidity while the oxygen produced increases alkalinity.

Alkaline pH favours conversion of starch to sugar.

Solute concentration increases inside guard cells, water is drawn into the cells by osmosis. Guard cells become turgid and the stoma opens.

At night when no photosynthesis, Respiration produces carbon (IV) oxide which raises acidity. This favours conversion of sugar to starch. Low sugar concentration leads to loss of turgidity in guard cells and stoma closes.

Explanation is based on accumulation of potassium ions

In day time (light) adenosine triphosphate (ATP) is produced which causes potassium ions to move into guard cells by active transport.

These ions cause an increase in solute concentration in guard cells that has been shown to cause movement of water into guard cells by osmosis.

Guard cells become turgid and the stoma opens.

At night potassium and chloride ions move out of the guard cells by diffusion and level of organic acid also decreases.

This causes a drop in solute concentration that leads to movement of water out of guard cells by osmosis.

Guard cells lose turgor and the stoma closes.

Process of Gaseous Exchange in Root Stem and Leaves of Aquatic and Terrestrial Plants

Gaseous Exchange in leaves of Terrestrial Plants

Gaseous exchange takes place by diffusion.

The structure of the leaf is adapted for gaseous exchange by having intercellular spaces that are filled.

These are many and large in the spongy mesophyll.

When stomata are open, carbon(IV) oxide from the atmosphere diffuses into the substomatal air chambers.

From here, it moves into the intercellular space in the spongy mesophyll layer.

The CO₂ goes into solution when it comes into contact with the cell surface and diffuses into the cytoplasm.

A concentration gradient is maintained between the cytoplasm of the cells and the intercellular spaces.

CO₂ therefore continues to diffuse into the cells.

The oxygen produced during photosynthesis moves out of the cells and into the intercellular spaces.

From here it moves to the substomatal air chambers and eventually diffuses out of the leaf through the stomata.

At night oxygen enters the cells while CO₂ moves out.

Gaseous exchange in the leaves of aquatic(floating) plants

Aquatic plants such as water lily have stomata only on the upper leaf surface.

The intercellular spaces in the leaf mesophyll are large.

Gaseous exchange occurs by diffusion just as in terrestrial plants.

Observation of internal structure of leaves of aquatic plants

Transverse section of leaves of an aquatic plant such as Nymphaea differs from that of terrestrial plant.

The following are some of the features that can be observed in the leave of an aquatic plant;

Absence of cuticle

Palisade mesophyll cells are very close to each other ie. compact.

Air spaces (aerenchyma) in spongy mesophyll are very large.

Sclereids (stone cells) are scattered in leaf surface and project into air spaces.

They strengthen the leaf making it firm and assist it to float.

Gaseous Exchange Through Stems Terrestrial Plants

Stems of woody plants have narrow openings or slits at intervals called **lenticels**.

They are surrounded by loosely arranged cells where the bark is broken.

They have many large air intercellular spaces through which gaseous exchange occurs.

Oxygen enters the cells by diffusion while carbon (IV) oxide leaves.

Unlike the rest of the bark, lenticels are permeable to gases and water.

Aquatic Plant Stems

The water lily, *Salvia* and *Wolffia* whose stems remain in water are permeable to air and water. Oxygen dissolved in the water diffuses through the stem into the cells and carbon (IV) oxide diffuses out into the water.

Gaseous Exchange in Roots Terrestrial Plants

Gaseous exchange occurs in the root hair of young terrestrial plants. Oxygen in the air spaces in the soil dissolves in the film of moisture surrounding soil particles and diffuses into the root hair along a concentration gradient. It diffuses from root hair cells into the cortex where it is used for respiration. Carbon (IV) oxide diffuses in the opposite direction. In older roots of woody plants, gaseous exchange takes place through lenticels.

Aquatic Plants

Roots of aquatic plants e.g. water lily are permeable to water and gases. Oxygen from the water diffuses into roots along a concentration gradient. Carbon (IV) oxide diffuses out of the roots and into the water. The roots have many small lateral branches to increase the surface area for gaseous exchange. They have air spaces that help the plants to float. Mangrove plants grow in permanently waterlogged soils, muddy beaches and at estuaries. They have roots that project above the ground level. These are known as breathing roots or pneumatophores. These have pores through which gaseous exchange takes place e.g. in *Avicennia* the tips of the roots have pores. Others have respiratory roots with large air spaces.

Gaseous Exchange in Animals

All animals take in oxygen for oxidation of organic compounds to provide energy for cellular activities. The carbon (IV) oxide produced as a by-product is harmful to cells and has to be constantly removed from the body. Most animals have structures that are adapted for taking in oxygen and for removal of carbon (IV) oxide from the body. These are called "respiratory organs". The process of taking in oxygen into the body and carbon (IV) oxide out of the body is called breathing or ventilation. Gaseous exchange involves passage of oxygen and carbon (IV) oxide through a respiratory surface by diffusion.

Types and Characteristics of Respiratory surfaces

Different animals have different respiratory surfaces.

The type depends mainly on the habitat of the animal, size, shape and whether body form is complex or simple.

Cell Membrane: In unicellular organisms the cell membrane serves as a respiratory surface.

Gills: Some aquatic animals have gills which may be external as in the tadpole or internal as in bony fish e.g. tilapia.

They are adapted for gaseous exchange in water.

Skin: Animals such as earthworm and tapeworm use the skin or body surface for gaseous exchange.

The skin of the frog is adapted for gaseous exchange both in water and on land.

The frog also uses **epithelium lining of the mouth or buccal cavity** for gaseous exchange.

Lungs: Mammals, birds and reptiles have lungs which are adapted for gaseous exchange.

Characteristics of Respiratory Surfaces

They are permeable to allow entry of gases.

They have a large surface area in order to increase diffusion.

They are usually thin in order to reduce the distance of diffusion.

They are moist to allow gases to dissolve.

They are well-supplied with blood to transport gases and maintain a concentration gradient.

Gaseous Exchange in Amoeba

Gaseous exchange occurs across the cell membrane by diffusion.

Oxygen diffuses in and carbon (IV) oxide diffuses out.

Oxygen is used in the cell for respiration making its concentration lower than that in the surrounding water.

Hence oxygen continually enters the cell along a concentration gradient.

Carbon (IV) oxide concentration inside the cell is higher than that in the surrounding water thus it continually diffuses out of the cell along a concentration gradient.

Gaseous Exchange in Insects

Gaseous exchange in insects e.g., grasshopper takes place across a system of tubes penetrating into the body known as the tracheal system.

The main trachea communicate with atmosphere through tiny pores called spiracles.

Spiracles are located at the sides of body segments;

Two pairs on the thoracic segments and eight pairs on the sides of abdominal segments.

Each spiracle lies in a cavity from which the trachea arises.

Spiracles are guarded with valves that close and thus prevent excessive loss of water vapour.

A filtering apparatus i.e. hairs also traps dust and parasites which would clog the trachea if they

gained entry.

The valves are operated by action of paired muscles. The main tracheae in the locust are located laterally along the length of the body on each side and they are interconnected across.

Each main trachea divides to form smaller tracheae, each of which branches into tiny tubes called tracheoles.

Each tracheole branches further to form a network that penetrates the tissues. Some tracheoles penetrate into cells in active tissue such as flight muscles.

These are referred to as intracellular tracheoles.

Tracheoles in between the cells are known as intercellular tracheoles.

The main tracheae are strengthened with rings of cuticle.

This helps them to remain open during expiration when air pressure is low.

Adaptation of Insect Tracheoles for Gaseous Exchange

The fine tracheoles are very thin about one micron in diameter in order to permeate tissue.

They are made up of a single epithelial layer and have no spiral thickening to allow diffusion of gases.

Terminal ends of the fine tracheoles are filled with a fluid in which gases dissolve to allow diffusion of oxygen into the cells.

Amount of fluid at the ends of fine tracheoles varies according to activity i.e. oxygen demand of the insect.

During flight, some of the fluid is withdrawn from the tracheoles such that oxygen reaches muscle cells faster and the rate of respiration is increased.

In some insects, tracheoles widen at certain places to form air sacs.

These are inflated or deflated to facilitate gaseous exchange as need arises.

Atmospheric air that dissolves in the fluid at the end of tracheoles has more oxygen than the surrounding cells of tracheole epithelium'.

Oxygen diffuses into these cells along a concentration gradient. '

Carbon (IV) oxide concentration inside the cells is higher than in the atmospheric .

Air diffuses out of the cells along a concentration gradient.

It is then removed with expired air.

Ventilation in Insects

Ventilation in insects is brought about by the contraction and relaxation of the abdominal muscles.

In locusts, air is drawn into the body through the thoracic spiracles and expelled through the abdominal spiracles.

Air enters and leaves the tracheae as abdominal muscles contract and relax.

The muscles contract laterally so the abdomen becomes wider and when they relax it becomes narrow.

Relaxation of muscles results in low pressure hence inspiration occurs while contraction of muscles results in higher air pressure and expiration occurs.

In locusts, air enters through spiracles in the thorax during inspiration and leaves through the abdominal spiracles during expiration.

This results in efficient ventilation.

Maximum extraction of oxygen from the air occurs sometimes when all spiracles close and hence contraction of abdominal muscles results in air circulating within the tracheoles.

The valves in the spiracles regulate the opening and closing of spiracles.

Observation of Spiracle in Locust

Some fresh grass is placed in a gas jar.

A locust is introduced into the jar.

A wire mesh is placed on top or muslin cloth tied around the mouth of the beaker with rubber band.

The insect is left to settle.

Students can approach and observe in silence the spiracles and the abdominal movements during breathing.

Alternatively the locust is held by the legs and observation of spiracles is made by the aid of hand lens.

Gaseous Exchange in Bony Fish (e.g, Tilapia)

Gaseous exchange in fish takes place between the gills and the surrounding water.

The gills are located in an opercular cavity covered by a flap of skin called the operculum.

Each gill consists of a number of thin leaf-like lamellae projecting from a skeletal base branchial arch (gill bar) situated in the wall of the pharynx.

There are four gills within the opercular cavity on each side of the head.

Each gill is made up of a bony gill arch which has a concave surface facing the mouth cavity (anterior) and a convex posterior surface.

Gill rakers are bony projections on the concave side that trap food and other solid particles which are swallowed instead of going over and damaging the gill filaments.

Two rows of gill filaments subtend from the convex surface. ***Adaptation of Gills for Gaseous Exchange***
Gill filaments are thin walled.

Gill filaments are very many (about seventy pairs on each gill), to increase surface area.

Each gill filament has very many gill lamellae that further increase surface area.

The gill filaments are served by a dense network of blood vessels that ensure efficient transport of gases.

It also ensures that a favourable diffusion gradient is maintained.

The direction of flow of blood in the gill lamellae is in the opposite direction to that of the water (counter current flow) to ensure maximum diffusion of gases.

Ventilation

As the fish opens the mouth, the floor of the mouth is lowered.

This increases the volume of the buccal cavity.

Pressure inside the mouth is lowered causing water to be drawn into the buccal cavity.

Meanwhile, the operculum is closed, preventing water from entering or leaving through the opening.

As the mouth closes and the floor of the mouth is raised, the volume of buccal cavity decreases while pressure in the opercular cavity increases due to contraction of opercular muscles.

The operculum is forced to open and water escapes.

As water passes over the gills, oxygen is absorbed and carbon dioxide from the gills dissolves in the water

As the water flows over the gill filaments oxygen in the water is at a higher concentration than that in the blood flowing, in the gill.

Oxygen diffuses through the thin walls of gill filaments/lamellae into the blood.

Carbon (IV) oxide is at a higher concentration in the blood than in the water.

It diffuses out of blood through walls of gill filaments into the water.

Counter Current Flow

In the bony fish direction of flow of water over the gills is opposite that of blood flow through the gill filaments .

This adaptation ensures that maximum amount of oxygen diffuses from the water into the blood in the gill filament.

This ensures efficient uptake of oxygen from the water.

Where the flow is along the same direction (parallel flow) less oxygen is extracted from the water.

Observation of Gills of a Bony Fish (Tilapia)

Gills of a fresh fish are removed and placed in a petri-dish with enough water to cover them.

A hand lens is used to view the gills.

Gill bar, gill rakers and two rows of gill filaments are observed.

Gaseous Exchange in an Amphibian - Frog

An adult frog lives on land but goes back into the water during the breeding season.

A frog uses three different respiratory surfaces.

These are the ***skin, buccal cavity and lungs***.

Skin

The skin is used both in water and on land.

It is quite efficient and accounts for 60% of the oxygen taken in while on land.

Adaptations of a Frog's Skin for Gaseous Exchange

The skin is a thin epithelium to allow fast diffusion.

The skin between the digits in the limbs (i.e. webbed feet) increase the surface area for gaseous exchange.

It is richly supplied with blood vessels for transport of respiratory gases.

The skin is kept moist by secretions from mucus glands.

This allows for respiratory gases to dissolve.

Oxygen dissolved in the film of moisture diffuses across the thin epithelium and into the blood which has a lower concentration of oxygen.

Carbon (IV) oxide diffuses from the blood across the skin to the atmosphere along the concentration gradient.

Buccal (Mouth) Cavity

Gaseous exchange takes place all the time across thin epithelium lining the mouth cavity.

Adaptations of Buccal Cavity for Gaseous Exchange

It has a thin epithelium lining the walls of the mouth cavity allowing fast diffusion of gases.
It is kept moist by secretions from the epithelium for dissolving respiratory gases.
It has a rich supply of blood vessels for efficient transport of respiratory gases.
The concentration of oxygen in the air within the mouth cavity is higher than that of the blood inside the blood vessels.
Oxygen, therefore dissolves in the moisture lining the mouth cavity and then diffuses into the blood through the thin epithelium.
On the other hand, carbon (IV) oxide diffuses in the opposite direction along a concentration gradient.

Lungs

There is a pair of small lungs used for gaseous exchange. Adaptation of Lungs

The lungs are thin walled for fast diffusion of gases.
Have internal foldings to increase surface area for gaseous exchange.
A rich supply of blood capillaries for efficient transport of gases.
Moisture lining for gases to dissolve.

Ventilation

Inspiration

During inspiration, the floor of the mouth is lowered and air is drawn in through the nostrils.
When the nostrils are closed and the floor of the mouth is raised, air is forced into the lungs.
Gaseous exchange occurs in the lungs, oxygen dissolves in the moisture lining of the lung and diffuses into the blood through the thin walls.
Carbon (IV) oxide diffuses from blood into the lung lumen.

Expiration

When the nostrils are closed and the floor of mouth is lowered by contraction of its muscles, volume of mouth cavity increases.
Abdominal organs press against the lungs and force air out of the lungs into buccal cavity.
Nostrils open and floor of the mouth is raised as its muscles relax.
Air is forced out through the nostrils.

Gaseous Exchange in a Mammal -Human

The breathing system of a mammal consists of a pair of **lungs** which are thin-walled elastic sacs lying in the thoracic cavity.
The thoracic cavity consists of vertebrae, sternum, ribs and intercostal muscles.
The thoracic cavity is separated from the abdominal cavity by the diaphragm.
The lungs lie within the thoracic cavity.
They are enclosed and protected by the ribs which are attached to the sternum and the thoracic vertebrae.
There are twelve pairs of ribs, the last two pairs are called 'floating ribs' because they are only attached to the vertebral column.

The ribs are attached to and covered by internal and external intercostal muscles.

The diaphragm at the floor of thoracic cavity consists of a muscle sheet at the periphery and a central circular fibrous tissue.

The muscles of the diaphragm are attached to the thorax wall.

The lungs communicate with the outside atmosphere through the bronchi, trachea, mouth and nasal cavities.

The trachea opens into the mouth cavity through the larynx.

A flap of muscle, the epiglottis, covers the opening into the trachea during swallowing.

This prevents entry of food into the trachea.

Nasal cavities are connected to the atmosphere through the external nares (or nostrils) which are lined with hairs and mucus that trap dust particles and bacteria, preventing them from entering into the lungs.

Nasal cavities are lined with cilia.

The mucus traps dust particles,

The cilia move the mucus up and out of the nasal cavities.

The mucus moistens air as it enters the nostrils.

Nasal cavities are winding and have many blood capillaries to increase surface area to ensure that the air is warmed as it passes along.

Each lung is surrounded by a space called the pleural cavity.

It allows for the changes in lung volume during breathing.

An internal pleural membrane covers the outside of each lung while an external pleural membrane lines the thoracic wall.

The pleural membranes secrete pleural fluid into the pleural cavity.

This fluid prevents friction between the lungs and the thoracic wall during breathing.

The trachea divides into two bronchi, each of which enters into each lung.

Trachea and bronchi are lined with rings of cartilage that prevent them from collapsing when air pressure is low.

Each bronchus divides into smaller tubes, the bronchioles.

Each bronchiole subdivides repeatedly into smaller tubes ending with fine bronchioles.

The fine bronchioles end in alveolar sacs, each of which gives rise to many alveoli.

Epithelium lining the inside of the trachea, bronchi and bronchioles has cilia and secretes mucus.

Adaptations of Alveolus to Gaseous Exchange

Each alveolus is surrounded by very many blood capillaries for efficient transport of respiratory gases.

There are very many alveoli that greatly increases the surface area for gaseous exchange.

The alveolus is thin walled for faster diffusion of respiratory gases.

The epithelium is moist for gases to dissolve.

Gaseous Exchange Between the Alveoli and the Capillaries

The walls of the alveoli and the capillaries are very thin and very close to each other.

Blood from the tissues has a high concentration of carbon (IV) oxide and very little oxygen compared to alveolar air.

The concentration gradient favours diffusion of carbon (IV) oxide into the alveolus and oxygen into the capillaries .

No gaseous exchange takes place in the trachea and bronchi.

These are referred to as dead space.

Ventilation

Exchange of air between the lungs and the outside is made possible by changes in the volumes of the thoracic cavity.

This volume is altered by the movement of the intercostal muscles and the diaphragm.

Inspiration

The ribs are raised upwards and outwards by the contraction of the external intercostal muscles, accompanied by the relaxation of internal intercostal muscles.

The diaphragm muscles contract and diaphragm moves downwards.

The volume of thoracic cavity increases, thus reducing the pressure.

Air rushes into the lungs from outside through the nostrils.

Expiration

The internal intercostal muscles contract while external ones relax and the ribs moved downwards and inwards.

The diaphragm muscles relaxes and it is pushed upwards by the abdominal organs. It thus assumes a dome shape.

The volume of the thoracic cavity decreases, thus increasing the pressure.

Air is forced out of the lungs.

As a result of gaseous exchange in the alveolus, expired air has different volumes of atmospheric gases as compared to inspired air.

Table 7.1: Comparison of Inspired and Expired Air (% by volume)

<i>Component</i>	<i>Inspired</i>	<i>Expired %</i>
Oxygen	21	16
Carbon	0.03	4
Nitrogen	79	79
Moisture	Variable	Saturated

Lung Capacity

The amount of air that human lungs can hold is known as lung capacity.

The lungs of an adult human are capable of holding 5,000 cm³ of air when fully inflated.

However, during normal breathing only about 500 cm³ of air is exchanged.

This is known as the tidal volume.

A small amount of air always remains in the lungs even after a forced expiration.

This is known as the residual volume.

The volume of air inspired or expired during forced breathing is called vital capacity.

Control of Rate Of Breathing

The rate of breathing is controlled by the respiratory centre in the medulla of the brain.

This centre sends impulses to the diaphragm through the phrenic nerve.

Impulses are also sent to the intercostal muscles.

The respiratory centre responds to the amount of carbon (IV) oxide in the blood.

If the amount of carbon (IV) oxide rises, the respiratory centre sends impulses to the diaphragm and the intercostal muscles which respond by contracting in order to increase the ventilation rate.

Carbon (IV) oxide is therefore removed at a faster rate.

Factors Affecting Rate of Breathing in Humans

Factors that cause a decrease or increase in energy demand directly affect rate of breathing.

Exercise, any muscular activity like digging.

Sickness

Emotions like anger, flight

Sleep.

Effects of Exercise on Rate of Breathing

Students to work in pairs.

One student stands still while the other counts (his/her) the number of breaths per minute.

The student whose breath has been taken runs on the sport vigorously for 10 minutes.

At the end of 10 minutes the number of breaths per minute is immediately counted and recorded.

It is noticed that the rate of breathing is much higher after exercise than at rest.

Dissection of a Small Mammal (Rabbit) to Show Respiratory Organs

The rabbit is placed in a bucket containing cotton wool which has been soaked in chloroform.

The bucket is covered tightly with a lid.

The dead rabbit is placed on the dissecting board ventral side upwards.

Pin the rabbit to the dissecting board by the legs.

Dissect the rabbit to expose the respiratory organs.

Ensure that you note the following features.

Ribs, intercostal muscles, diaphragm, lungs, bronchi, trachea, pleural membranes, thoracic cavity.

DISEASES OF RESPIRATORY SYSTEM

Asthma is a chronic disease characterised by narrowing of air passages. Causes:

Allergy

Due to pollen, dust, fur, animal hair, spores among others.

If these substances are inhaled, they trigger release of chemical substances and they may cause swelling of the bronchioles and bring about an asthma attack.

Heredity

Asthma is usually associated with certain disorders which tend to occur in more than one member of a given family, thus suggesting a hereditary tendency.

Emotional or mental stress

Strains the body immune system hence predisposes to asthma attack.

Symptoms

Asthma is characterized by wheezing and difficulty in breathing accompanied by feeling of tightness in the chest as a result of contraction of the smooth muscles lining the air passages.

Treatment and Control

There is no definite cure for asthma.

The best way where applicable is to avoid whatever triggers an attack (allergen).

Treatment is usually by administering drugs called bronchodilators.

The drugs are inhaled, taken orally or injected intravenously depending on severity of attack to relieve bronchial spasms.

Bronchitis

This is an inflammation of bronchial tubes.

Causes

This is due to an infection of bronchi and bronchioles by bacteria and viruses.

Symptoms

Difficulty in breathing.

Cough that produces mucus.

Treatment

Antibiotics are administered.

Pulmonary Tuberculosis

Tuberculosis is a contagious disease that results in destruction of the lung tissue.

Causes

Tuberculosis is caused by the bacterium ***Mycobacterium tuberculosis***.

Human tuberculosis is spread through droplet infection i.e., in saliva and sputum.

Tuberculosis can also spread from cattle to man through contaminated milk.

From a mother suffering from the disease to a baby through breast feeding.

The disease is currently on the rise due to the lowered immunity in persons with HIV and AIDS

Tuberculosis is common in areas where there is dirt, overcrowding and malnourishment.

Symptoms

It is characterised by a dry cough, lack of breath and body wasting.

Prevention

Proper nutrition with a diet rich in proteins and vitamins to boost immunity.

Isolation of sick persons reduces its spread.

Utensils used by the sick should be sterilised by boiling.

Avoidance of crowded places and living in well ventilated houses.

Immunisation with B.C.G. vaccine gives protection against tuberculosis.

This is done a few days after birth with subsequent boosters.

Treatment

Treatment is by use of antibiotics.

Pneumonia

Pneumonia is infection resulting in inflammation of lungs.

The alveoli get filled with fluid and bacterial cells decreasing surface area for gaseous exchange.

Pneumonia is caused by bacteria and virus.

More infections occur during cold weather.

The old and the weak in health are most vulnerable.

Symptoms

Pain in the chest accompanied by a fever, high body temperatures (39-40°C) and general body weakness.

Prevention

Maintain good health through proper feeding.

Avoid extreme cold.

Treatment

If the condition is caused by pneumococcus bacteria, antibiotics are administered.

If breathing is difficult, oxygen may be given using an oxygen mask.

Whooping Cough

Whooping cough is an acute infection of respiratory tract.

The disease is more common in children under the age of five but adults may also be affected.

Causes

It is caused by *Bordetella pertusis* bacteria and is usually spread by droplets produced when a sick person coughs.

Symptoms:

Severe coughing and frequent vomiting.

Thick sticky mucus is produced.

Severe broncho-pneumonia.

Convulsions in some cases.

Prevention

Children may be immunised against whooping cough by means of a vaccine which is usually combined with those against diphtheria and tetanus.

It is called "Triple Vaccine" or ***Diphtheria, Pertusis*** and ***Tetanus*** (DPT).

Treatment

Antibiotics are administered.

To reduce the coughing, the patient should be given drugs.

3.6 Transport in plants

TRANSPORT IN PLANTS

Introduction

Transport is the movement of substances within an organism.

All living cells require oxygen and food for various metabolic processes.

These substances must be transported to the cells.

Metabolic processes in the cells produce excretory products which should be eliminated before they accumulate.

The excretory products should be transported to sites of excretion.

Organisms like amoeba are unicellular.

They have a large surface area to volume ratio.

The body is in contact with the environment.

Diffusion is adequate to transport substances across the cell membrane and within the organism.

Large multi-cellular organisms have complex structure where cells are far from each other hence diffusion alone cannot meet the demand for supply and removal of substances.

Therefore an elaborate transport system is necessary.

Transport in plants

- Simple plants such as mosses and liverworts lack specialized transport system.
- Higher plants have specialized transport systems known as the vascular bundle.
- Xylem transports water and mineral salts .
- Phloem transports dissolved food substances like sugars.

Internal structure of roots and root hairs

The main functions of roots are ;

- Anchorage
- absorption.
- storage
- gaseous exchange.

The outermost layer in a root is the piliferous layer.

This is a special epidermis of young roots whose cells give rise to root hairs.

Root hairs are microscopic outgrowths of epidermal cells.

They are found just behind the root tip,

They are one cell thick for efficient absorption of substances.

They are numerous and elongated providing a large surface area for absorption of water and mineral salts.

Root hairs penetrate the soil and make close contact with it.

Below the piliferous layer is the cortex.

This is made up of loosely packed, thin walled parenchyma cells.

Water molecules pass through this tissue to reach the vascular bundles.

In some young plant stems, cortex cells contain chloroplasts.

The endodermis (starch sheath) is a single layer of cells with starch grains.

The endodermis has a Casparian strip which has an impervious deposit controlling the entry of water and mineral salts into xylem vessels.

Pericycle forms a layer next to the endodermis.

Next to the pericycle is the vascular tissue.

In the Dicotyledonous root, xylem forms a star shape in the centre, with phloem in between the arms.

It has no pith. In monocotyledonous root, xylem alternates with phloem and there is a pith in the centre.

Internal structure of a root hair cell

The Stem

The main functions of the stem are;

support and exposure of leaves and flowers to the environment,

conducting water and mineral salts

conducting manufactured food from leaves to other parts of the plant.

In monocotyledonous stems, vascular bundles are scattered all over the stem, while in dicotyledonous stems vascular bundles are arranged in a ring.

Vascular bundles are continuous from root to stems and leaves.

The epidermis forms a single layer of cells enclosing other tissues.

The outer walls of the cells have a waxy cuticle to prevent excessive loss of water.

The cortex is a layer next to the epidermis.

It has collenchyma, parenchyma and sclerenchyma cells ***Collenchyma***

Is next to the epidermis and has thickened walls at the corners which strengthen the stem.

Parenchyma

Cells are irregular in shape, thin walled and loosely arranged hence creating intercellular spaces filled with air.

They are packing tissues and food storage areas.

Sclerenchyma

Cells are closely connected to vascular bundles.

These cells are thickened by deposition of lignin and they provide support to plants. ***Pith***

Is the central region having parenchyma cells.

Absorption of Water and Mineral Salts Absorption of Water

Root hair cell has solutes in the vacuole and hence a higher osmotic pressure than the surrounding soil water solution.

Water moves into the root hair cells by osmosis along a concentration gradient.

This makes the sap in the root hair cell to have a lower osmotic pressure than the surrounding

cells.

Therefore water moves from root hair cells into the surrounding cortex cells by osmosis.

The process continues until the water gets into the xylem vessels .

Uptake of Mineral Salts

If the concentration of mineral salts in solution is greater than its concentration in root hair cell, the mineral salts enter the root hair cell by diffusion.

If the concentration of mineral salts in the root hair cells is greater than in the soil water, the mineral salts enter the root hairs by active transport.

Most minerals are absorbed in this way.

Mineral salts move from cell to cell by active transport until they reach the xylem vessel.

Once inside the xylem vessels, mineral salts are transported in solution as the water moves up due to root pressure, capillary attraction and cohesion and adhesion forces.

Transpiration

Transpiration is the process by which plants lose water in the form of water vapour into the atmosphere.

Water is lost through ***stomata, cuticle and lenticels***.

Stomatal transpiration:

This accounts for 80-90% of the total transpiration in plants.

Stomata are found on the leaves.

Cuticular transpiration: The cuticle is found on the leaves, and a little water is lost through it. Plants with thick cuticles do not lose water through the cuticle.

Lenticular transpiration

Is loss' of water through lenticels.

These are found on stems of woody plants.

Water lost through the stomata and cuticle by evaporation leads to evaporation of water from surfaces of mesophyll cells .

The mesophyll cells draw water from the xylem vessels by osmosis.

The xylem in the leaf is continuous with xylem in the stem and root.

Structure and function of Xylem

Movement of water is through the xylem.

Xylem tissue is made up of vessels and tracheids.

Xylem Vessels

Xylem vessels are formed from cells that are elongated along the vertical axis and arranged end to end.

During development, the cross walls and organelles disappear and a continuous tube is formed.

The cells are dead and their walls are strengthened by deposition of lignin.

The lignin has been deposited in various ways.

This results in different types of thickening

Annular.

Simple spiral.

Double spiral.

Reticulate.

The bordered pits are areas without lignin on xylem vessels and allow passage of water in and out of the lumen to neighbouring cells.

Tracheids

Tracheids have cross-walls that are perforated.

Their walls are deposited with lignin.

Unlike the xylem vessels, their end walls are tapering or chisel-shaped.

Their lumen is narrower.

Besides transport of water, xylem has another function of strengthening the plant which is provided by xylem fibres and xylem parenchyma.

Xylem fibres ;

Are cells that are strengthened with lignin.

They form wood.

Xylem parenchyma:

These are cells found between vessels.

They form the packing tissue.

Forces involved in Transportation of Water and Mineral Salts

Transpiration pull

As water vaporises from spongy mesophyll cells into sub-stomatal air spaces, the cells of mesophyll cells develop a higher osmotic pressure than adjacent cells.

Water is then drawn into mesophyll cells by osmosis from adjacent cells and finally from xylem vessels.

A force is created in the leaves which pulls water from xylem vessels in the stem and root.

This force is called ***transpiration pull***.

Cohesion and Adhesion:

The attraction between water molecules is called cohesion.

The attraction between water molecules and the walls of xylem vessels is called adhesion.

The forces of cohesion and adhesion maintain a continuous flow of water in the xylem from the root to the leaves.

Capillarity:

Is the ability of water to rise in fine capillary tubes due to surface tension.

Xylem vessels are narrow, so water moves through them by capillarity.

Root Pressure:

If the stem of a plant is cut above the ground level, it is observed that cell sap continues to come out of the cut surface.

This shows that there is a force in the roots that pushes water up to the stem.

This force is known as root pressure.

Importance of Transpiration

Transpiration leads to excessive loss of water if unchecked.

Some beneficial effects are

Replacement of water lost during the process.

Movement of water up the plant is by continuous absorption of water from the soil.

Mineral salts are transported up the plant.

Transpiration ensures cooling of the plant in hot weather.

Excessive loss of water leads to wilting' and eventually death if water is not available in the soil.

Factors Affecting Transpiration

The factors that affect transpiration are grouped into two.

i.e. environmental and structural.

Environmental factors Temperature

High temperature increases the internal temperature of the leaf. which in turn increases kinetic energy of water molecules which increases evaporation.

High temperatures dry the air around the leaf surface maintaining a high concentration gradient.

More water vapour is therefore lost from the leaf to the air.

Humidity

The higher the humidity of the air around the leaf, the lower the rate of transpiration.

The humidity difference between the inside of the leaf and the outside is called the saturation deficit.

In dry atmosphere, the saturation deficit is high.

At such times, transpiration rate is high.

Wind

Wind carries away water vapour as fast as it diffuses out of the leaves.

This prevents the air around the leaves from becoming saturated with vapour.

On a windy day, the rate of transpiration is high.

Light Intensity

When light intensity is high; more stomata open hence high rate of transpiration.

Atmospheric Pressure

The lower the atmospheric pressure the higher the kinetic energy of water molecules hence more evaporation.

Most of the plants at higher altitudes where atmospheric pressure is very low have adaptations to prevent excessive water-loss.

Availability of Water

The more water there is in the soil, the more is absorbed by the plant and hence a lot of water is lost by transpiration.

Structural Factors

Cuticle

Plants growing in arid or semi-arid areas have leaves covered with a thick wax cuticle.

Stomata

The more the stomata, the higher the rate of transpiration.

Xerophytes have few stomata which reduce water-loss.

Some have sunken stomata which reduces the rate of transpiration as the water vapour accumulates in the pits.

Others have stomata on the lower leaf surface hence reducing the rate of water-loss.

Some plants have reversed stomatal rhythm whereby stomata close during the day and open at night.

This helps to reduce water-loss.

Leaf size and shape

Plants in wet areas have large surface area for transpiration.

Xerophytes have small narrow leaves to reduce water-loss.

The photometer can be used to determine transpiration in different environmental conditions.

Translocation of organic compounds

Translocation of soluble organic products of photosynthesis within a plant is called translocation.

It occurs in phloem in sieve tubes.

Substances translocated include glucose, amino acids, vitamins.

These are translocated to the growing regions like stem, root apex, storage organs e.g. corms, bulbs and secretory organs such as nectar glands.

Phloem

phloem is made up of;

sieve tubes,

companion cells

parenchyma, a packing tissue

sclerenchyma, a strengthening tissue

Sieve Tubes

These are elongated cells arranged end to end along the vertical axis.

The cross walls are perforated by many pores to make a sieve plate.

Most organelles disappear and those that remain are pushed to the sides of the sievetube.

Cytoplasmic strands pass through the pores in the plate into adjacent cells.

Food substances are translocated through cytoplasmic strands.

Companion Cells

Companion cells are small cells with large nuclei and many mitochondria.

They are found alongside each sieve element.

The companion cell is connected to the tube through plasmodesmata.

The mitochondria generate energy required for translocation.

Phloem Parenchyma

These are parenchyma cells between sieve elements.

They act as packing tissue.

3.7 TRANSPORT IN ANIMALS

The Circulatory System

Large and complex animals have circulatory systems that consist of tubes, a transport fluid and a means of pumping the fluid.

Blood is the transport fluid which contains dissolved substances and cells.

The tubes are blood vessels through which dissolved substances are circulated around the body.

The heart is the pumping organ which keeps the blood in circulation.

The types of circulatory system exist in animals: open and closed.

In an open circulatory system;

The heart pumps blood into vessels which open into body spaces known as a haemocoel.

Blood comes into contact with tissues.

A closed circulatory system;

Found in vertebrates and annelids where the blood is confined within blood vessels and does not come into direct contact with tissues.

Transport in Insects

In an insect, there is a tubular heart just above the alimentary canal.

This heart is suspended in a pericardial cavity by ligaments.

The heart has five chambers and extends along the thorax and abdomen.

Blood is pumped forwards into the aorta by waves of contractions in the heart.

It enters the haemocoel and flows towards the posterior.

The blood flows back into the heart through openings in each chamber called ostia.

The ostia have valves which prevent the backflow of blood.

Blood is not used as a medium for transport of oxygen in insects.

This is because oxygen is supplied directly to the tissues by the tracheal system.

The main functions of blood in an insect are to transport nutrients, excretory products and hormones.

Mammalian Circulatory System

Mammals have a closed circulatory system where a powerful heart pumps blood into arteries.

The arteries divide into smaller vessels called arterioles.

Each arteriole divides to form a network of capillaries inside the tissues.

The capillaries eventually re-unite to form venules, which form larger vessels called veins.

The veins take the blood back to the heart.

Blood from the heart goes through the pulmonary artery to the lungs and then back to the heart through pulmonary vein.

This circulation is called pulmonary circulation

Oxygenated blood leaves the heart through the aorta and goes to all the tissues of the body.

From the tissues, deoxygenated blood flows back to the heart through the vena cava.

This circulation is called systemic circulation.

In each complete circulation, the blood flows into the heart twice.

This is called double circulation.

Some other animals like fish have a single circulation.

Blood flows only once through the heart for every complete circuit.

Structure and Function of the Heart

The heart has four chambers:

Two atria (auricles) and two ventricles.

The left and right side of the heart are separated by a muscle wall (septum) so that oxygenated and deoxygenated blood does not mix.

Deoxygenated blood from the rest of the body enters the heart through the vena cava

.

Blood enters the right atrium, then through tricuspid valve into right ventricle.

Then via semi-lunar valve to the pulmonary artery to the lungs.

Oxygenated blood from the lungs enters the heart through pulmonary vein.

It enters the left atrium of the heart, then through bicuspid valve into left ventricle.

Then via semi-lunar valves to aorta which takes oxygenated blood round the body.

A branch of the aorta called coronary artery supplies blood to the heart muscle.

The coronary vein carries blood from the heart muscle to the pulmonary artery which then takes it to the lungs for oxygenation.

The Heartbeat

The heart is capable of contracting and relaxing rhythmically without fatigue due to its special muscles called cardiac muscles.

The rhythmic contraction of the heart arises from within the heart muscles without nervous stimulation.

The contraction is said to be myogenic.

The heartbeat is initiated by the pacemaker or sino-atrio-node (SAN) which is located in the right atrium.

The wave of excitation spreads over the walls of atria.

It is picked up by the atrio-ventricular node which is located at the junction:

Of the atria and ventricles, from where the Purkinje tissue spreads the wave to the walls of the ventricles.

The heart contracts and relaxes rhythmically at an average rate of 72 times per minute.

The rate of the heartbeat is increased by the sympathetic nerve, while it is slowed down by the vagus nerve.

Heartbeat is also affected by hormones e.g. adrenaline raises the heartbeat.

Structure and Function of Arteries, Capillaries and Veins

Arteries

Arteries carry blood away from the heart.

They carry oxygenated blood except pulmonary artery which carries deoxygenated blood to the lungs.

Arteries have a thick, muscular wall, which has elastic and collagen fibres that resist the pressure of the blood flowing in them.

The high pressure is due to the pumping action of the heart.

The pressure in the arteries originates from the pumping action of the heart.

The pulse or number of times the heart beats per minute can be detected by applying pressure on an artery next to the bone.

e.g. by placing the finger/thumb on the wrist.

The innermost layer of the artery is called endothelium which is smooth.

It offers least possible resistance to blood flow.

Have a narrow lumen .

The aorta forms branches which supply blood to all parts of the body.

These arteries divide into arterioles which further divide to form capillaries.

Capillaries

Capillaries are small vessels whose walls are made of endothelium which is one cell thick.

This provides a short distance for exchange of substances.

Capillaries penetrate tissues,

The lumen is narrow therefore blood flowing in capillaries is under high pressure.

Pressure forces water and dissolved substances out of the blood to form tissue fluid.

Exchange of substances occurs between cells and tissue fluid. Part of the tissue fluid passes back into capillaries at the venule end.

Excess fluid drains into small channels called lymph capillaries which empty their contents into lymphatic vessels.

Capillaries join to form larger vessels called venules which in turn join to form veins which transport blood back to the heart.

Veins

Veins carry deoxygenated blood from the tissues to the heart (except pulmonary vein which carries oxygenated blood from the lungs to the heart).

Veins have a wider lumen than arteries.

Their walls are thinner than those of arteries.

Blood pressure in the veins is low.

Forward flow of blood in veins is assisted by contraction of skeletal muscles, hence the need for exercise.

Veins have valves along their length to prevent backflow of blood.

This ensures that blood flows towards the heart.

The way the valves work can be demonstrated on the arm.

By pressing on one vein with two fingers, leaving one and pushing blood toward the heart then releasing the latter finger, it can be observed that the part in between is left with the vein not being visible.

This is because blood does not flow back towards the first finger.

Diseases and Defects of Circulatory System Thrombosis

Formation of a clot in the blood vessels is called thrombosis.

Coronary thrombosis is the most common.

It is caused by blockage of coronary artery which supplies blood to the heart.

Blockage may be due to artery becoming fibrous or accumulation of fatty material on the artery walls.

Narrow coronary artery results in less blood reaching the heart muscles.

A serious blockage can result in heart attack which can be fatal.

Heavy intake of fat, alcohol, being overweight and emotional stress can cause coronary thrombosis.

A blockage in the brain can lead to a stroke causing paralysis of part of the body, coma or even death.

A healthy lifestyle, avoiding a lot of fat in meals and avoiding alcohol can control the

disease.

Arteriosclerosis

This condition results from the inner walls having materials being deposited there or growth of fibrous connective tissue.

This leads to thickening of the wall of the artery and loss of elasticity.

Normal blood flow is hindered.

Arteriosclerosis can lead to thrombosis or hypertension.

A person with hypertension which is also called high blood pressure has his/her blood being pumped more forcefully through the narrow vessels.

This puts stress on the walls of the heart and arteries.

Regular exercise, healthy diet and avoiding smoking can help maintain normal blood pressure.

Varicose Veins

Superficial veins especially at the back of the legs become swollen and flabby due to some valves failing to function properly.

This results to retention of tissue fluid.

Regular physical exercise will prevent this condition.

Repair of valves through surgery can also be done.

Wearing surgical stockings may ease a mild occurrence.

Structure and Function of Blood

Composition of Blood

The mammalian blood is made up of a fluid medium called plasma with substances dissolved in it.

Cellular components suspended in plasma include;

erythrocytes (red blood cells),

leucocytes (white blood cells)

thrombocytes (platelets)

blood proteins.

Plasma

This is a pale yellow fluid consisting of 90% water.

There are dissolved substances which include;

glucose, amino acids, lipids, salts,

hormones, urea, fibrinogen, albumen,

antibodies, some enzymes suspended cells.

Serum is blood from which fibrinogen and cells have been removed.

The functions of plasma include:

Transport of red blood cells which carry oxygen.

Transport dissolved food substances round the body.

Transport metabolic wastes like nitrogenous wastes and carbon (IV) oxide in solution about 85% of the carbon (IV) oxide is carried in form of hydrogen carbonates.

Transport hormones from sites of production to target organs.

Regulation of pH of body fluids.

Distributes heat round the body hence regulate body temperature.

Absence of nucleus leaves room for more haemoglobin to be packed in the cell to enable it to carry more oxygen.

Haemoglobin contained in red blood cells is responsible for the transport of oxygen.

Haemoglobin + Oxygen = oxyhaemoglobin

(Hb) + (4O₂) → (HbO₄)

Erythrocytes (Red Blood Cells)

In humans these cells are circular biconcave discs without nuclei. Oxygen is carried in form of oxyhaemoglobin.

Haemoglobin readily picks up oxygen in the lungs where concentration of oxygen is high.

In the tissues, the oxyhaemoglobin breaks down (dissociates) easily into haemoglobin and oxygen.

Oxygen diffuses out of the red blood cells into the tissues.

Haemoglobin is then free to pick up more oxygen molecules.

The biconcave shape increases their surface area over which gaseous exchange takes place.

Due to their ability, they are able to change their shape to enable themselves squeeze inside the narrow capillaries.

There are about five million red blood cells per cubic millimetre of blood.

They are made in the bone marrow of the short bones like sternum, ribs and vertebrae.

In the embryo they are made in the liver and spleen.

Erythrocytes have a life span of about three to four months after which they are destroyed in the liver and spleen.

Also in the red blood cells is carbonic anhydrase which assists in the transport of carbon (IV) oxide.

Leucocytes (White Blood Cells)

These white blood cells have a nucleus.

They are divided into two:

Granulocytes (also phagocytes or polymorphs)

Agranulocytes .

White blood cells defend the body against disease.

Neutrophils form 70% of the granulocytes.

Others are eosinophils and basophils.

About 24% agranulocytes are called lymphocytes, while 4% agranulocytes are monocytes.

The leucocytes are capable of amoebic movement.

They squeeze between the cells of the capillary wall to enter the intercellular spaces.

They engulf and digest disease causing organisms (pathogens) by phagocytosis.

Some white blood cells may die in the process of phagocytosis.

The dead phagocytes, dead organisms and damaged tissues form pus.

Lymphocytes produce antibodies which inactivate antigens.

Antibodies include:

Antitoxins which neutralise toxins.

Agglutinins cause bacteria to clump together and they die.

Lysins digest cell membranes of microorganisms.

Opsonins adhere to outer walls of microorganisms making it easier for phagocytes to ingest them.

Lymphocytes are made in the thymus gland and lymph nodes.

There are about 7,000 leucocytes per cubic millimetre of blood.

Platelets (Thrombocytes)

Platelets are small irregularly shaped cells formed from large bone marrow cells called megakaryocytes.

There are about 250,000 platelets per cubic millimetre of blood.

They initiate the process of blood clotting.

The process of clotting involves a series of complex reactions whereby fibrinogen is converted into a fibrin clot.

When blood vessels are injured platelets are exposed to air and they release

thromboplastin which initiates the blood clotting process.

Thromboplastin neutralises heparin the anti-clotting factor in blood and activates prothrombin to thrombin.

The process requires calcium ions and vitamin K.

Thrombin activates the conversion of fibrinogen to fibrin which forms a meshwork of fibres on the cut surface to trap red blood cells to form a clot.

The clot forms a scab that stops bleeding and protects the damaged tissues from entry of micro-organisms.

Blood clotting reduces loss of blood when blood vessels are injured.

Excessive loss of blood leads to anaemia and dehydration.

Mineral salts lost in blood leads to osmotic imbalance in the body.

This can be corrected through blood transfusion and intravenous fluid.

ABO Blood Groups

There are four types of blood groups in human beings: A, B, AB and O.

These are based on types of proteins on the cell membrane of red blood cells.

There are two types of proteins denoted by the letters A and B which are antigens.

In the plasma are antibodies specific to these antigens denoted as *a* and *b*.

A person of blood group A has A antigens on the red blood cells and *b* antibodies in plasma.

A person of blood group B has B antigens on red blood cells and *a* antibodies in plasma.

A person of blood group AB has A and B antigens on red blood cells and no antibodies in plasma.

A person of blood group o has no antigens on red blood cells and *a* and *b* antibodies in plasma.

Blood groups

Blood	Antigens	Antibodi
A	A	B
B	B	a
AB	A and B	None
o	None	a and b

Blood Transfusion

Blood transfusion is the transfer of blood from a donor to the circulatory system of the recipient.

A recipient will receive blood from a donor if the recipient has no corresponding antibodies to the donor's antigens.

If the donor's blood and the recipient's blood are not compatible, agglutination occurs whereby red blood cells clump together.

Blood typing

A person of blood group O can donate blood to a person of any other blood group.

A person of blood group O is called a universal donor.

A person of blood group AB can receive blood from any other group.

A person with blood group AB is called a universal recipient.

A person of blood group A can only donate blood to another person with blood group A or a person with blood group AB.

A person of blood group B can only donate blood to somebody with blood group B or a person with blood group AB.

A person with blood group AB can only donate blood to a person with blood group AB.

Blood screening has become a very important step in controlling HIV/AIDS.

It is therefore important to properly screen blood before any transfusion is done.

Rhesus Factor

The Rhesus factor is present in individuals with the Rhesus antigen in their red blood cells.

Such individuals are said to be Rhesus positive (Rh+), while those without the antigen are Rhesus negative (Rh-).

If blood from an Rh+ individual is introduced into a person who is Rh- , the latter develops antibodies against the Rhesus factor.

There may not be any reaction after this transfusion.

However a subsequent transfusion with Rh+ blood causes a severe reaction, and agglutination occurs i.e. clumping of red blood cells.

The clump can block the flow of blood, and cause death.

Erythroblastosis foetalis (haemolytic disease of the newborn) results when an Rh-mother carries an Rh+ foetus.

This arises when the father is Rh+.

During the latter stage of pregnancy, fragments of Rhesus positive red blood cells of the foetus may enter mother's circulation.

These cause the mother to produce Rhesus antibodies which can pass across the placenta to the foetus and destroy foetal red blood cells.

During the first pregnancy, enough antibodies are not formed to affect the foetus.

Subsequent pregnancies result in rapid production of Rhesus antibodies by the mother.

These destroy the red blood cells of the foetus, the condition called haemolytic disease of the newborn.

The baby is born anaemic and with yellow eyes (jaundiced).

The condition can be corrected by a complete replacement of baby's blood with safe healthy blood.

Lymphatic System

The lymphatic system consists of lymph vessels.

Lymph vessels have valves to ensure unidirectional movement of lymph.

Lymph is excess tissue fluid i.e. blood minus blood cells and plasma proteins.

Flow of lymph is assisted by breathing and muscular contractions.

Swellings called lymph glands occur at certain points along the lymph vessels.

Lymph glands are oval bodies consisting of connective tissues and lymph spaces.

The lymph spaces contain lymphocytes which are phagocytic.

Lymph has the same composition as blood except that it does not contain red blood cells and plasma proteins.

Lymph is excess tissue fluid.

Excess tissue fluid is drained into lymph vessels by hydrostatic pressure.

The lymph vessels unite to form major lymphatic system.

The main lymph vessels empty the contents into sub-clavian veins which take it to the heart.

Immune Responses

Immune response is the production of antibodies in response to antigens.

An antigen is any foreign material or organism that is introduced into the body and causes the production of antibodies.

Antigens are protein in nature.

An antibody is a protein whose structure is complementary to the antigen.

This means that a specific antibody deals with a specific antigen to make it harmless.

When harmful organisms or proteins invade the body, lymphocytes produce complementary antibodies, while bone marrow and thymus gland produce more phagocytes and lymphocytes respectively.

Types of Immunity

There are two types of **immunity; natural and artificial.**

Natural Immunity is also called innate immunity.

It is inherited from parent to offspring.

Artificial Immunity can be natural or induced.

When attacked by diseases like chicken pox, measles and mumps, those who recover from these diseases develop resistance to any subsequent infections of the same diseases.

This is natural acquired immunity.

Artificial Acquired Immunity:

When attenuated (weakened) or dead microorganisms are introduced into a healthy person.

The lymphocytes synthesise the antibodies which are released into the lymph and eventually reach the blood.

The antibodies destroy the invading organisms.

The body retains 'memory' of the structure of antigen.

Rapid response is ensured in subsequent infections.

Vaccines generally contain attenuated disease causing organisms.

Artificial Passive Acquired Immunity:

Serum containing antibodies is obtained from another organism, and confers immunity for a short duration.

Such immunity is said to be passive because the body is not activated to produce the antibodies.

Importance of Vaccination

A vaccine is made of attenuated, dead or nonvirulent micro-organism that stimulate cells in the immune system to recognise and attack disease causing agent through production of antibodies.

Vaccination protects individuals from infections of many diseases like smallpox, tuberculosis and poliomyelitis.

Diseases like smallpox, tuberculosis and tetanus were killer diseases but this is no longer the case.

Diphtheria Pertussis Tetanus (DPT) vaccine protects children against diphtheria, whooping cough and tetanus.

Bacille Calmette Guérin (BCG) vaccine is injected at birth to children to protect them against tuberculosis.

Measles used to be a killer disease but today, a vaccine injected into children at the age of nine months prevents it.

At birth children are given an inoculation through the mouth of the poliomyelitis vaccine.

Allergic Reactions

An allergy is a hypersensitive reaction to an antigen by the body.

The antibody reacts with the antigen violently.

People with allergies are oversensitive to foreign materials like dust, pollen grains, some foods, some drugs and some air pollutants.

Allergic reactions lead to production of **histamine** by the body.

Histamine causes swelling and pain.

Allergic reactions can be controlled by avoiding the allergen and administration of anti-histamine drugs.

3.8 NUTRITION IN PLANTS AND ANIMALS

Nutrition in Plants (Autotrophisms)

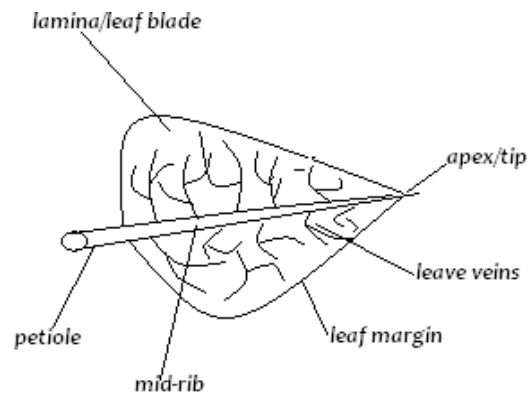
Structure of the Leaf

Structure

The external structure of the leaf consists of a leaf stalk or petiole and a broad leafblade or lamina.

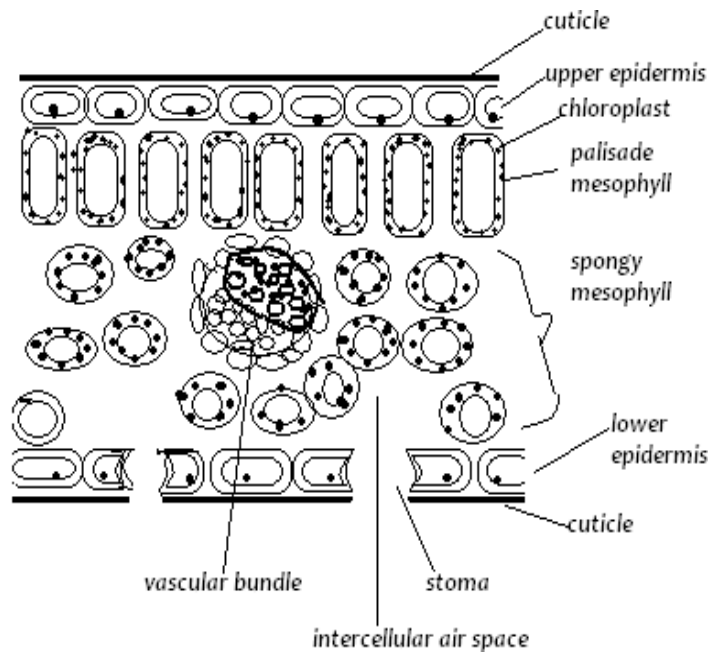
The lamina has a main vein midrib from which smaller veins originate.

The outline of the leaf is the margin and the tip forms the apex.



External structure of a leaf

Internal Structure of the Leaf



Transvers section of a leaf showing internal structure

Epidermis

This is the outer layer of cells, normally one cell thick.

It is found in both the upper and lower leaf surfaces.

The cells are arranged end to end.

The epidermis offers protection and maintains the shape of the leaf.

It is covered by a layer of cuticle which reduces evaporation.

Leaf Mesophyll

Consists of the palisade layer, next to upper epidermis, and the spongy layer next to the lower epidermis.

Palisade Mesophyll Layer

The cells are elongated and arranged close to each other leaving narrow air spaces. These contain numerous chloroplasts and are the main photosynthetic cells.

In most plants, the chloroplasts are distributed fairly uniformly throughout the cytoplasm. In certain plants growing in shaded habitats in dim light, most chloroplasts migrate to the upper region of the palisade cells in order to maximise absorption of the limited light available.

Spongy Mesophyll Layer

The cells are spherical in shape.

They are loosely arranged, with large intercellular spaces between them.

The spaces are air-filled and are linked to the stomatal pores.

The spongy mesophyll cells have fewer chloroplasts than the palisade mesophyll cells.

Vascular Bundles

These are made up of the xylem and the phloem tissues.

The xylem transports water and mineral salts to the leaves.

The phloem transports food manufactured in the leaf to the other parts of the plant and from storage organs to other parts.

Adaptations of Leaf for Photosynthesis

Presence of veins with vascular bundles. Xylem vessels transport water for photosynthesis.

Phloem transports manufactured food from leaves to other parts of the plant.

Leaf lamina is thin to allow for penetration of light over short distance to reach photosynthetic cells.

Broad lamina provides a large surface area for absorption of light and carbon (IV) oxide.

Transparent cuticle and epidermal layer allow light to penetrate to mesophyll cells.

Palisade cells are close to the upper epidermis for maximum light absorption.

Presence of numerous chloroplasts in palisade mesophyll traps maximum light.

Chloroplast contain chlorophyll that traps light energy.

Spongy mesophyll layer has large intercellular air spaces allowing for gaseous exchange.

Presence of stomata for efficient gaseous exchange (entry of carbon (IV) oxide into leaf and exit of oxygen).

Mosaic arrangement of leaves to ensure no overlapping of leaves hence every leaf is exposed to light.

Structure and Function of Chloroplasts

Chloroplasts are large organelles (5 μm in diameter) found in the cytoplasm of green plant cells.

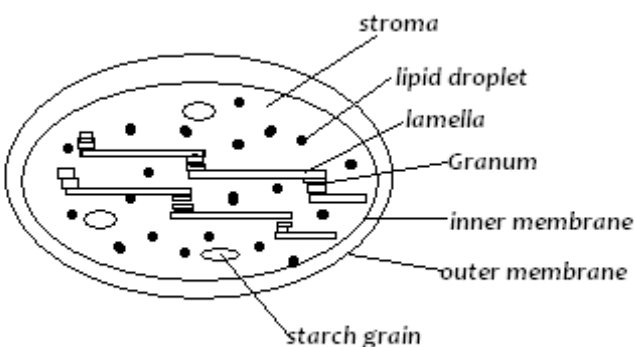
They are visible under the light microscope.

They contain chlorophyll, a green pigment and other carotenoids which are yellow, orange and red in colour.

Certain plants have red or purple leaves due to abundance of these other pigments.

Chlorophyll absorbs light energy and transforms it into chemical energy.

The other pigments absorb light but only to pass it onto chlorophyll.



Structure of Chloroplast

The wall of chloroplast consists of an outer and an inner membrane.

The two make up the chloroplast envelope.

Inner membrane encloses a system of membranes called lamellae.

At intervals, the membranes form stacks of fluid filled sacs known as grana (singular granum).

Chloroplast and other pigments are attached to the grana. In between the lamellae is a gel-like stroma, that contains starch grains and lipid droplets.

Enzymes for the dark stage reaction (light independent stage) are embedded in the stroma.

Enzymes for the light dependent stage occur in the grana.

Functions

.Absorption of light by chlorophyll and other pigments.

Light stage of photosynthesis occurs on the grana. (transformation of light energy to chemical energy.)

Carbon fixation to form carbohydrate takes place in the stroma which has enzymes for dark stage of photosynthesis.

Process of Photosynthesis

Photosynthesis involves a series of chemical reactions, all of which take place inside chloroplasts.

A general equation for photosynthesis is:

Carbon (IV) Oxide + Water $\xrightarrow{\text{light energy}}$ Glucose + Oxygen
chlorophyll

$6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow[\text{chlorophyll}]{\text{light}}$ $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

The reaction occurs in two main phases or stages.

The initial state requires light and it is called the light dependent stage or simply light stage.

It takes place on the lamellae surfaces.

Its products are used in the dark stage.

The dark stage does not require light although it occurs in the light and is called light independent stage.

Factors Influencing Photosynthesis

Certain factors must be provided for before photosynthesis can take place. The rate or amount of photosynthesis is also influenced by the quantity or quality of these same factors.

Carbon(IV) Oxide Concentration

Carbon (IV) oxide is one of the raw materials for photosynthesis.

No starch is formed when leaves are enclosed in an atmosphere without carbon (IV) oxide.

The concentration of carbon (IV) oxide in the atmosphere remains fairly constant at about 0.03% by volume.

However, it is possible to vary the carbon (IV) oxide concentration under experimental conditions.

Increasing the carbon (IV) oxide concentration up to 0.1 % increases the rate of photosynthesis. Further increase reduces the rate.

Light Intensity

Light supplies the energy for photosynthesis.

Plants kept in the dark do not form starch.

Generally, increase in light intensity up to a certain optimum, increases the rate of photosynthesis.

The optimum depends on the habitat of the plant.

Plants that grow in shady places have a lower optimum than those that grow in sunny places. **Water**

Water is necessary as a raw material for photosynthesis.

The amount of water available greatly affects the rate of photosynthesis.

The more water available, the more the photosynthetic rate, hence amount of food made.

Effect of water on photosynthesis can only be inferred from the yield of crops.

It is the main determinant of yield (limiting factor in the tropics).

Temperature

The reactions involved in photosynthesis are catalysed by a series of enzymes.

A suitable temperature is therefore necessary.

The optimum temperature for photosynthesis in most plants is around 30°C.

This depends on the natural habitat of the plant.

Some plants in temperate regions have 20°C as their optimum while others in the tropics have 45°C as their optimum temperature.

The rate of photosynthesis decreases with a decrease in temperature below the optimum.

In most plants, photosynthesis stops when temperatures approach 0°C although some arctic plant species can photosynthesise at -2°C or even -3°C.

Likewise, increase in temperature above the optimum decreases the rate and finally the reactions stop at temperatures above 40°C due to enzyme denaturation.

However, certain algae that live in hot springs e.g. *Oscillatoria* can photosynthesise at 75°C

Chlorophyll

Chlorophyll traps or harnesses the energy from light.

Leaves without chlorophyll do not form starch.☐

Nutrition in Animals (Heterotrophism)

Meaning and Types of Heterotrophism

This is a mode of nutrition whereby organisms feed on complex organic matter from other plants or animals.

All animals are heterotrophs.

Their mode of feeding is also said to be holozoic to distinguish it from other special types of heterotrophic nutrition namely:

- saprophytism
- parasitism.
- Saprophytism/saprotrophism- occurs in most fungi and some forms of bacteria.

Saprophytes feed on dead organic matter and cause its decomposition or decay.

Parasitism is a mode of feeding whereby one organism called the parasite feeds on or lives in another organism called the host and harms it.

Modes of Feeding in Animals

Animals have developed various structures to capture and ingest food.

The type of structures present depend on the method of feeding and the type of food.

Carnivorous animals feed on whole animals or portions of their flesh.

Herbivorous animals feed on plant material.

Omnivorous animals feed on both plants and animal materials.

Feeding in Mammals

The jaws and teeth of mammals are modified according to the type of food eaten.

Mammals have different kinds of teeth.

Each type of teeth has a particular role to play in the feeding process.

Feeding in Mammals

- The jaws and teeth of mammals are modified according to the type of food eaten.
- Mammals have different kinds of teeth.
- Each type of teeth has a particular role to play in the feeding process.
- This condition is described as heterodont.
- The teeth of reptiles and amphibians are all similar in shape and carry out the same function.
- They are said to be homodont.

Digestive System and Digestion in Humans

Organs that are involved with feeding in humans constitute the digestive system.

Digestive System and Associated Glands

Human digestive system starts at the mouth and ends at the anus.

This is the alimentary canal.

Digestion takes place inside the lumen of the alimentary canal. The epithelial wall that faces the lumen has mucus glands (goblet cells).

These secrete mucus that lubricate food and prevent the wall from being digested by digestive enzymes.

Present at specific regions are glands that secrete digestive enzymes.

The liver and pancreas are organs that are closely associated with the alimentary canal.

Their secretions get into the lumen and assist in digestions.

Digestive system consists of:

Mouth.

Oesophagus.

Stomach.

Small intestines - consist of duodenum, the first part next to the stomach, ileum - the last part that ends up in a vestigial caecum and appendix which are nonfunctional.

Large intestines consist of: colon and rectum that ends in the anus.

Ingestion, Digestion and Absorption

Feeding in humans involves the following processes:

Ingestion: This is the introduction of the food into the mouth.

Digestion: This is the mechanical and chemical breakdown of the food into simpler, soluble and absorbable units.

Absorption: Taking into blood the digested products.

Assimilation: Use of food in body cells.

Mechanical breakdown of the food takes place with the help of the teeth.

Chemical digestion involves enzymes.

Digestion in the Mouth

In the mouth, both mechanical and chemical digestion takes place.

Food is mixed with saliva and is broken into smaller particles by the action of teeth.

Saliva contains the enzyme amylase.

It also contains water and mucus which lubricate and soften food in order to make swallowing easy.

Saliva is slightly alkaline and thus provides a suitable pH for amylase to act on cooked starch, changing it to maltose.

The food is then swallowed in the form of semisolid balls known as boluses.

Each bolus moves down the oesophagus by a process known as peristalsis.

Circular and longitudinal muscles along the wall of the alimentary canal contract and relax pushing the food along.

Digestion in the Stomach

In the stomach, the food is mixed with gastric juice secreted by gastric glands in the stomach wall.

Gastric juice contains pepsin, rennin and hydrochloric acid.

The acid provides a low pH of 1.5-2.0 suitable for the action of pepsin.

Pepsin breaks down protein into peptides.

Rennin coagulates the milk protein casein. The stomach wall has strong circular and longitudinal muscles whose contraction mixes the food with digestive juices in the stomach.

Digestion in the Duodenum

In the duodenum the food is mixed with bile and pancreatic juice.

Bile contains bile salts and bile pigments.

The salts emulsify fats, thus providing a large surface area for action of lipase.

Pancreatic juice contains three enzymes:

Trypsin which breaks down proteins into peptides and amino acids,

Amylase which breaks down starch into maltose, and

Lipase which breaks down lipids into fatty acids and glycerol.

These enzymes act best in an alkaline medium which is provided for by the bile.

Digestion in ileum

Epithelial cells in ileum secrete intestinal juice, also known as succus entericus.

This contains enzymes which complete the digestion of protein into amino acids, carbohydrates into monosaccharides and lipids into fatty acids and glycerol.

Absorption

This is the diffusion of the products of digestion into the blood of the animal.

It takes place mainly in the small intestines though alcohol and some glucose are absorbed in the stomach.

The ileum is adapted for absorption in the following ways:

It is highly coiled.

The coiling ensures that food moves along slowly to allow time for its digestion and absorption.

It is long to provide a large surface area for absorption.

The epithelium has many finger-like projections called villi (singular villus).

They greatly increase the surface area for absorption.

Villi have microvilli that further increase the surface area for absorption.

The wall of villi has thin epithelial lining to facilitate fast diffusion of products of digestion.

Has numerous blood vessels for transport of the end products of digestion.

Has lacteal vessels; for absorption of fatty acids and glycerol and transport of lipids.

Types of Mammalian Teeth

- Mammals have four kinds of teeth.
- The incisors are found at the front of the jaw.
- They are sharp-edged and are used for biting.
- The canines are located at the sides of the jaw.
- They are pointed and are used for tearing and piercing.
- The premolars are next to the canines and the molars are at the back of the jaw.
- Both premolars and molars are used for crushing and grinding.
- Teeth are replaced only once in a lifetime.
- The first set is the milk or deciduous teeth.
- These are replaced by the second set or the permanent teeth.



canine

incisor

premolar

molar

Different types of teeth in humans

Dentition refers to the type of teeth, the number and their arrangement in the jaw.

A dental formula shows the type and number of teeth in each half of the jaw.

The number of teeth in half of the upper jaw is represented above a line and those on the lower jaw below the line.

The first letter of each type of teeth is used in the formula i.e. i = incisors, c = canines, pm = premolars and m = molars.

The total number is obtained by multiplying by two (for the two halves of each jaw).

Adaptation of Teeth to Feeding

In general, incisors are for cutting, canines for tearing while premolars and molars are for grinding.

However, specific modifications are observed in different mammals as an adaptation to the type of food they eat.

Teeth of Herbivores

Incisors are long and flat with a sharp chisel-like edge for cutting.

The enamel coating is thicker in front than at the back so that as the tooth wears out, a sharp edge is maintained.

Canines are reduced or absent.

If absent, the space left is called the diastema.

The diastema allows the tongue to hold food and push it to the grinding teeth at the back of the

mouth.

Premolars and molars:

These are transversely ridged.

The ridges on the upper teeth fit into grooves on the lower ones.

This gives a sideways grinding surface.

The teeth of herbivores have open roots i.e., wide opening into the pulp cavity. This ensures a continued adequate supply of food and oxygen to the tooth.

In some herbivores, such as rabbits and elephants, the incisors continue to grow throughout life.

Teeth of Carnivores

Incisors are reduced in size and pointed.

They are well suited for grasping food and holding prey.

Canines are long, pointed and curved.

They are used for piercing and tearing flesh as well as for attack and defence.

Premolars and molars: In general, they are long and longitudinally ridged to increase surface area for crushing.

Carnassial Teeth: These are the last premolars on the upper jaw and the first molars on the lower one.

They are enlarged for cutting flesh.

They act as a pair of shears.

They also crush bones.

The teeth of carnivores have closed roots i.e., only a very small opening of the pulp cavity to allow food and oxygen to keep teeth alive.

Once broken, no re-growth can take place.

Teeth of Omnivores

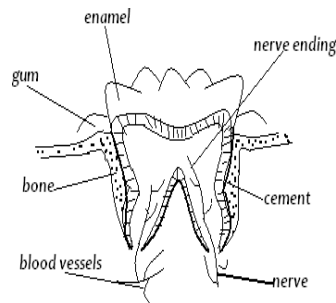
Incisors have a wide surface for cutting.

Canines are bluntly pointed for tearing.

Premolars and molars have cusps for crushing and grinding.

The premolars have two blunt cusps while the molars have three to four.

Internal Structure of tooth



Internal structure of a tooth

The tooth consists of two main parts:

Crown: The portion above the gum; it is covered by the enamel.

Root: The portion below the gum; it is covered by the cement.

The tooth has two roots.

Neck: Is the region at the same level with the gum.

It forms the junction between the crown and the root.

It is covered by enamel. Incisors and canines have one root only.

Premolars have one or two roots while molars have two to three roots each.

Internally, the bulk of the tooth is made up of dentine which consists of living cells and extends to the root.

It is composed of calcium salts, collagen and water.

It is harder than bone but wears out with use.

This is why it is covered by enamel which is the hardest substance in a mammal's body.

Pulp Cavity: Contains blood vessels which provide nutrients to the dentine and remove waste products.

It also contains nerve endings which detect heat, cold and pain.

Cement: Fixes the tooth firmly to the jaw bone.

Common Dental Diseases

Dental Carries

Dental carries are the holes or cavities that are formed as acid corrodes enamel and eventually the dentine.

Causes

This is caused by bacteria acting on the food left between teeth and on the cusp.

Acids are formed that eventually corrode the enamel.

The pulp cavity is eventually reached.

A lot of pain is experienced then.

The bacteria then infect the pulp cavity and the whole tooth decays.

Treatment

Treatment depends on the extent of the dental caries:

Extraction of Tooth.

Filling - this involves replacing the dentine with amalgam, a mixture of hard elements e.g. silver and tin.

Root Canal Treatment - This involves surgery and reconstruction.

It saves severely damaged teeth.

The nerves in the root canal are surgically severed.

The tooth is cleaned and filled up with amalgam.

Periodontal Diseases

These are diseases of the gum.

The gum becomes inflamed, and starts bleeding.

Progression of the disease leads to infection of the fibres in the periodontal membranes and the tooth becomes loose.

This condition is known as pyorrhoea.

The diseases are caused by poor cleaning of the teeth.

The accumulation of food particles leading to formation of plaque, lack of adequate vitamin A and C in the diet.

Treatment

Nutrition - by taking adequate balanced diet rich in vitamins A and C.

Antibiotics are used to kill bacteria.

Anti-inflammatory drugs are given.

Antiseptic is prescribed to use in cleaning the mouth daily to prevent further proliferation of bacteria.

The plaque is removed-drilled away - a procedure known as scaling. In order to maintain healthy teeth the following points should be observed:

A proper diet that includes calcium and vitamins, particularly vitamin D is essential.

The diet should also contain very small quantities of fluorine to strengthen the enamel.

Large quantities of fluorine are harmful.

The enamel becomes brown, a condition known as dental fluorosis.

Chewing of hard fibrous foods like carrots and sugar cane to strengthen and cleanse the teeth.

Proper use of teeth e.g. not using teeth to open bottles and cut thread.

Regular and thorough brushing of teeth after meals.

Dental floss can be used to clean between the teeth.

Not eating sweets and sugary foods between meals.

Regular visits to the dentist for checkup.

Washing the mouth with strong salt solution or with any other mouth wash with antiseptic properties.

3.9 REPRODUCTION IN PLANTS

Asexual Reproduction

Asexual reproduction is the formation of offspring from a single parent.

The offspring are identical to the parent.

Types of asexual reproduction.

Binary fission in amoeba.

Spore formation in *Rhizopus*.

Budding in yeast.

Binary fission

This involves the division of the parent organism into two daughter cells.

The nucleus first divides into two and then the cytoplasm separates into two portions

Binary fission also occurs in bacteria, *Paramecium*, *Trypanosoma* and *Euglena*.

Spore formation in Rhizopus

Rhizopus is a saprophytic fungus which grows on various substrate such as bread, rotting fruits or other decaying organic matter.

The vegetative body is called mycelium which has many branched threads called hyphae.

Horizontal hyphae are called stolons.

Vertical hyphae are called sporangiophore.

The tips of sporangiophore become swollen to form sporangia, the spore bearing structure.

Each sporangium contains many spores.

As it matures and ripens, it turns black in colour.

When fully mature the sporangium wall bursts and releases spores which are dispersed by wind or insects.

When spores land on moist substratum, they germinate and grow into a new *Rhizopus* and start another generation.

Spore formation in ferns

The fern plant is called a sporophyte.

On the lower side of the mature leaves are sori (Singular: sorus) which bear spores.

Budding in Yeast

Budding involves the formation of a protrusion called a bud from the body of the organism.

The bud separates from the parent cell, in yeast budding goes on so fast and the first bud starts to form another bud before the separation.

A short chain or mass of cells is formed.

Sexual Reproduction in Plants

In flowering plants, the flower is the reproductive organ which is a specialised shoot consisting of a modified stem and leaves.

The stem-like part is the pedicel and receptacle, while modified leaves form corolla and calyx.

Structure of a flower

A typical flower consists of the following parts:

Calyx –

made up of sepals.

They enclose and protect the flower when it is in a bud. Some flowers have an outer whorl made of sepal-like structures called epicalyx.

Corolla –

consists of petals. The petals are brightly coloured in insect - pollinated flowers.

Androecium –

Is the male part of the flower. It consists of stamens.

Each stamen consists of a filament whose end has an anther.

Inside the anther are pollen sacs which contain pollen grains.

Gynoecium (pistil) –

Is the female part of the flower.

Each carpel consists of an ovary, a style and a stigma.

The ovary contains ovules which become seeds after fertilisation.

A ***monocarpous pistil*** has one carpel e.g. beans.

A ***polycarpous pistil*** has many carpels.

If the carpels are free, it is called ***apocarpous*** as in rose and *Bryophyllum*,

In carpels that are fused it is called ***syncarpous*** as in *Hibiscus*.

A ***complete flower*** has all the four floral parts.

A regular flower can be divided into two halves by any vertical section passing through the centre. e.g. morning glory.

Irregular flower can be divided into two halves in only one plane e.g. crotalaria.

Pollination

This is the transfer of pollen grains from the anther to the stigma.

Types of pollination

Self pollination is the transfer of pollen grains from the anther of one flower to the stigma of the same flower.

Cross-pollination is the transfer of pollen grains from the anther of one flower to the stigma of a

different flower, of the same species.

Agents of pollination

Agents of pollination include wind, insects, birds and mammals.

Insect pollinators include bees, butterflies and mosquitoes.

Mechanisms that hinder self-pollination

Stamens ripen early and release their pollen grains before the stigma, mature. This is called protandry e.g. in sunflower.

The stigma matures earlier and dries before the anthers release the pollen grains.

This is called protogyny and is common in grasses.

Self sterility or incompatibility

Pollen grains are sterile to the stigma of the same flower, e.g. in maize flower.

Shorter stamens than pistils.

Fertilisation in Plants

The pollen grain contains the generative nucleus and a tube nucleus.

When the pollen grain lands on the stigma, it absorbs nutrient and germinates forming a pollen tube.

This pollen tube grows through the style pushing its way between the cells.

It gets nourishment from these cells.

The tube nucleus occupies the position at the tip of the growing pollen tube.

The generative nucleus follows behind the tube nucleus, and divides to form two male gamete nuclei.

The pollen tube enters the ovule through the micropyle.

When the pollen tube penetrates the ovule disintegrates and the pollen tube bursts open leaving a clear way for the male nuclei.

One male nucleus fuses with the egg cell nucleus to form a diploid zygote which develops into an embryo.

The other male gamete nucleus fuses with the polar nucleus to form a triploid nucleus which forms the primary endosperm.

This is called double fertilisation.

After fertilisation the following changes take place in a flower:

- The integuments develop into seed coat (testa).
- The zygote develops into an embryo.
- The triploid nucleus develops into an endosperm.
- The ovules become seeds.
- The ovary develops into a fruit.
- The ovary wall develops into pericarp.
- The style, dries up and falls off leaving a scar.
- The corolla, calyx and stamens dry up and fall off.

- In some the calyx persists.

Fruit formation

Fruit development without fertilisation is called parthenocarpy

e.g. as in pineapples and bananas.

Such fruits do not have seeds.

Classification of fruits

False fruits develop from other parts such as calyx, corolla and receptacle,

e.g. apple and pineapple which develop from an inflorescence.

True fruits develop from the ovary, e.g. bean fruit (pod).

True fruits can be divided into fleshy or succulent fruits e.g. berries and drupes and dryfruits.

The dry ones can be divided into Dehiscent which split open to release seeds and indehiscent which do not open.

Placentation

This is the arrangement of the ovules in an ovary.

Marginal placentation:

The placenta appears as one ridge on the ovary wall e.g. bean.

Parietal placentation:

The placenta is on the ridges on ovary wall.

Ovules are in them e.g. pawpaw.

Axile placentation:

The placenta is in the centre.

Ovary is divided into a number of loculi. e.g. orange.

Basal placentation.

The placenta is formed at the base of the ovary e.g. sunflower.

Free Central placentation.

Placenta is in the centre of the ovary.

There are no loculi e.g. in primrose.

Methods of fruit and seed dispersal

Fleshy fruits are eaten by animals.

Animals are attracted to the fruits by the bright colour, scent or the fact that it is edible.

The seeds pass through the digestive tract undamaged and are passed out with faeces.

E.g. tomatoes and guavas.

Such seeds have hard, resistant seed coats.

Others have fruits with hooks or spines that stick on animal fur or on clothes.

Later the seeds are brushed off or fall off on their own e.g. *Bidens pilosa* (Black jack).

Wind dispersal

Fruits and seeds are small and light in order to be carried by air currents.

A fruit that is a capsule e.g. tobacco split or has pores at the top e.g. Mexican poppy.

The capsule is attached to a long stalk when swayed by wind the seeds are released and scattered.

Some seeds have hairy or feather-like structures which increase their surface area so that they can be blown off by the wind e.g. *Sonchus*.

Others have wing-like structures e.g. Jacaranda and Nandi Flame.

These extensions increase the surface area of fruits and seeds such that they are carried by the wind.

Water dispersal

Fruits like coconut have fibrous mesocarp which is spongy to trap air, the trapped air makes the fruit light and buoyant to float on water.

Plants like water lily produce seeds whose seed coats trap air bubbles.

The air bubbles make the seeds float on water and are carried away.

The pericarp and seed coat are waterproof.

Self-dispersal (explosive) Mechanism

This is seen in pods like bean and pea.

Pressure inside the pod forces it to open along lines of weakness throwing seeds away from parent plant.

3.10 REPRODUCTION IN ANIMALS

Sexual reproduction involves the fusion of gametes.

In animals two individuals are involved, a male and a female.

Special organs known as gonads produce gametes.

In males testes produce sperms while in females ovaries produce ova.

The fusion of male gamete and female gamete to form a zygote is called fertilisation.

There are two types of fertilisation. External and internal. Example in amphibians takes place in water.

The male mounts the female and shed sperms on the eggs as they are laid.

Eggs are covered by slippery jelly-like substance which provides protection.

Many eggs are released to increase the chances of survival.

Internal fertilisation

This occurs in reptiles, birds and mammals.

Fertilisation occurs within the body of the female.

Fewer eggs are produced because there are higher chances of fertilisation since sperms are released into the female body.

Reproduction in Humans

Structure of female reproduction system

The female reproduction system consist of the following:

Ovaries

Are two oval cream coloured structures found in lower abdomen below the kidneys.

Oviducts.

They produce the ova.

Are tubes which conduct the ova produced by the ovaries to the uterus.

Fertilisation occurs in the upper part of the oviduct.

Uterus

The uterus is a hollow muscular organ found in the lower abdomen.

The embryo develops inside the uterus.

The inner lining endometrium supplies nutrients to embryo.

The embryo is implanted into the inner uterine wall- the endometrium which nourishes the embryo.

The thick muscles of the uterus assist in parturition.

Cervix

Has a ring of muscles that separates the uterus from the vagina.

It forms the opening to the uterus

Vagina

Is a tube that opens to the outside and it acts as the copulatory and birth canal through the vulva.

Structure of male reproductive system

The male reproductive system consists of the following

Testis:

Each testis is a mass of numerous coiled tubes called seminiferous tubules.

Each is enclosed within a scrotal sac that suspends them between the thighs.

This ensures that sperms are maintained at a temperature lower than that of the main body.

Seminiferous tubules

The lining of seminiferous tubules consists of actively dividing cells which give rise to sperms.

Between the seminiferous tubules are interstitial cells which produce the male hormones called androgens e.g. testosterone.

The seminiferous tubules unite to form the epididymis, which is a coiled tube where sperms are stored temporarily.

Vas deferens (sperm duct) is the tube through which sperms are carried from testis to urethra.

Seminal vesicle produces an alkaline secretion which nourishes the spermatozoa.

Prostate gland

Produces an alkaline secretion to neutralise vaginal fluids.

Cowpers' gland

Secretes an alkaline fluid.

All these fluids together with spermatozoa form semen.

Urethra

Is a long tube through which the semen is conducted during copulation.

It also removes urine from the bladder.

Penis

Is an intro-mittent organ which is inserted into the vagina during copulation.

Fertilisation in Animals

Fertilisation is preceded by copulation in which the erect penis is inserted into the vagina.

This leads to ejaculation of semen.

The sperms swim through the female's genital tract to the upper part of the oviduct.

The head of the sperm penetrates the egg after the acrosome releases lytic enzymes to dissolve the egg membrane.

The tail is left behind.

Sperm nucleus fuses with that of the ovum and a zygote is formed.

A fertilisation membrane forms around the zygote which prevents other sperms from penetrating the zygote.

implantation

After fertilisation the zygote begins to divide mitotically as it moves towards the uterus.

It becomes embedded in the wall of the uterus a process called implantation.

By this time the zygote is a hollow ball of cells called blastocyst or embryo.

In the uterus the embryo develops villi which project into uterus for nourishment later the villi and endometrium develop into placenta.

Embryonic membranes

Embryonic membranes develop around the embryo.

The outermost membrane is the chorion which forms the finger-like projections (chorionic villi) which supply nutrients to the embryo.

The amnion surrounds the embryo forming a fluid filled cavity within which the embryo lies.

Amniotic cavity is filled with amniotic fluid.

This fluid acts as a shock absorber and protects the foetus against mechanical injury.

It also regulates temperature.

The chorionic villi, allantois together with the endometrium form the placenta.

The embryo is attached to the placenta by a tube called umbilical cord which has umbilical vein and artery.

The maternal blood in the placenta flows in the spaces lacuna and surrounds capillaries from umbilical vein and artery.

The umbilical cord increases in length as the embryo develops.

Role of placenta

Protection

Maternal blood and foetal blood do not mix.

This ensures that the pathogens and toxins from maternal blood do not reach the foetus.

The placenta allows maternal antibodies to pass into the foetus, providing the foetus with immunity.

Nutrition

The placenta facilitates the transfer of nutrients from maternal blood to foetus.

Excretion

Placenta facilitates the removal of nitrogenous wastes from the foetus' blood to maternal blood.

Gaseous exchange

Oxygen from the maternal blood diffuses into the foetal blood while carbon (IV) oxide from foetal blood diffuses into maternal blood.

Production of hormones

Placenta produces progesterone and oestrogen.

Gestation period

The period between conception and birth is called gestation.

In humans gestation takes nine months (40 weeks).

The embryo differentiates into tissues and organs during this period.

Week 1 to 3:

Zygote divides to form blastocyst.

Implantation takes place.

The three germ layers form endoderm, mesoderm and ectoderm.

Nervous system starts to form.

Week 4 to 7:

Development of circulating and digestive systems.

Further development of nervous system, formation of sensory organs,

All major internal organs are developed.

At week 5, heartbeat starts.

Week 8 to 24:

All organs well developed including sex organs.

Hair, finger and toe nails grow.

Foetus moves and eyelids open.

Week 25- 30:

The fully developed foetus responds to touch and noises and moves vigorously.

The head turns and faces downwards ready for birth.

Week 31-40:

Foetus increases in size.

Birth occurs.

Reproductive Hormones

<i>Hormone</i>	<i>Source</i>	<i>Functions</i>
Follicle Stimulating Hormone (FSH)	Pituitary gland	Development of ovarian follicle; stimulates secretion of oestrogen by the ovary
Luteinising Hormone	Pituitary gland	Causes ovulation; causes development of Graafian follicle into the corpus luteum; causes secretion of progesterone by the ovary
Prolactin	Pituitary gland	Initiates production and secretion of milk by the mammary glands
Oxytocin	Pituitary gland	Causes contraction of the uterus during parturition (birth)
Progesterone	Corpus luteum in the ovary	Causes contraction of wall of the uterus to thicken after ovulation
Oestrogen	Ovary	Causes changes in the uterine wall in preparation for implantation; initiates development of secondary sexual characteristics
Androgens-Testosterone	Interstitial cells of Testis	Stimulates the development of secondary sexual Characteristics
Interstitial Cell Hormone (ICSH)	Pituitary gland	Stimulates the interstitial cells of testis to release Androgens
Human Chorionic Gonadotrophin (HCG)	Chorionic villi	Stops the degeneration of the corpus luteum for production of oestrogen and progesterone

Secondary Sexual Characteristics

Male

- Testosterone is the main androgen that stimulates the development of secondary sexual characteristics.
- Broadening of the shoulders.
- Deepening of the voice due to enlargement of larynx.
- Hair at the pubic area, armpit and chin regions.
- Penis and testis enlarge and produce sperms.
- Body becomes more masculine.

Female

- Enlargement of mammary glands.
- Hair grows around pubic and armpit regions.
- Widening of the hips.
- Ovaries mature and start producing ova.
- Menstruation starts.
- Oestrogen triggers the onset of secondary sexual characteristics.

Sexually transmitted infections (STI)

<i>Disease</i>	<i>Causative</i>	<i>Method of Transmission</i>	<i>Symptoms</i>	<i>Prevention/control</i>
Gonorrhoea	Bacterium Neisseria Gonorrhoea	-Sexual contact - during birth for Infants -Sharing towels	-Itching of urethra -yellowish discharge pain as males urinate, vaginal discharge. with odour in females	A void indiscriminate sex. Treat both partners infected A void sharing linen
Syphilis	Bacterium Treponema Palladium	-Sexual contact - During birth for infants. - Sharing towels and linen	Solitary painless ulcer-on genital or mucous -Rashes, muscles and papules on hands, feet lips, genital areas	Treat at primary infection stage -Avoid indiscriminate sex. - A void sharing linen

Trichomoniasis	Protozoan Trichomonas Vaginalis	-Sexual contact -contaminated linen, underwear and toilet seats	Itching of urethra or vagina in females, smelly, yellow Discharge	-Avoid sharing linen -Avoid indiscriminate sex -personal hygiene
Hepatitis	Virus Hepatitis B	-Sexual contact -blood contaminated needles and syringes	Fever, nausea, jaundice, loss of appetite, yellow Urine	-Avoid indiscriminate sex -use disposable needles and syringes - strict personal hygiene
Candidiasis	Fungus Albicans	-spread through sexual contact - sharing linen and towels	Itching and burning sensation and white discharge from Genitals	-Avoid indiscriminate sex - Treat both partners
Herpes (Simplex)	Virus Herpes	-sexual contact kissing, contaminated needles	Lesions on skin and mucous membranes of buccal cavity vagina or head of Penis	- A void indiscriminate sex and contaminated needles and syringes.
HIV and Aids	Virus Human Deficiency	-sexual contact -blood -contaminated instruments -Through breast milk and body fluids. -Through birth canal for infants	-chronic diarrhoea -weight loss (more than 10% body weight lost in a month) - constant, persistent cough, skin infectious (herpes zoster)	-Avoid indiscriminate sex. -Use screened blood - No sharing of tooth brushes, razors - Use disposable needles

Menstrual Cycle

This is characterized by discharge of blood and tissue debris (menses) from the uterus every 28 days.

This is due to the breakdown of the endometrium which occurs when the level of progesterone falls and the girl starts to menstruate.

The follicle stimulating hormone (FSH) causes the Graafian follicle to develop and also stimulate the

ovary to release oestrogen.

Oestrogen hormone triggers the onset of secondary sexual characteristics.

Luteinising hormone (L.H) causes the mature ovum to be released from the Graafian follicle - a process called ovulation.

After ovulation progesterone hormone is produced.

After menstruation, the anterior lobe of the pituitary gland starts secreting the follicle stimulating hormone (FSH) which causes the Graafian follicle to develop in the ovary.

It also stimulates the ovary tissues to secrete oestrogen.

Oestrogen brings about the repair and healing of the inner lining of the uterus (endometrium) which had been destroyed during menstruation.

Oestrogen level stimulates the pituitary gland to produce (Luteinising Hormone (L.H)).

This hormone makes the mature Graafian follicle to release the ovum into the funnel of oviduct, a process called ovulation.

After releasing the ovum, the Graafian follicle changes into a yellow body called corpus luteum.

The luteinising hormone stimulates the corpus luteum to secrete a hormone called progesterone which stimulates the thickening and vascularisation of endometrium.

This prepares the uterine wall for implantation of the blastocyst.

If fertilisation takes place, the level of progesterone increases and thus inhibits FSH from stimulating the maturation of another Graafian follicle.

If fertilisation does not occur, the corpus luteum disintegrates and the level of progesterone goes down.

The endometrium, sloughs off and menstruation occurs.

Advantages of Reproduction Asexual

Good qualities from parents are retained in the offspring without variation.

New individuals produced asexually mature faster.

Process does not depend on external factors which may fail such as pollination.

New individuals obtain nourishment from parent and so are able to survive temporarily under unsuitable conditions.

No indiscriminate spreading of individuals which can result in wastage of offspring.

Takes a shorter time and leads to rapid colonization.

Disadvantages of asexual reproduction

New offspring may carry undesirable qualities from parents.

Offspring may be unable to withstand changing environmental conditions.

Faster maturity can cause overcrowding and stiff competition.

Reduced strength and vigour of successive generations.

Advantages of sexual reproduction

Leads to variations.

Variations which are desirable often show hybrid vigour.

High adaptability of individuals to changing environmental conditions.

Variations provide a basis for evolutionary changes.

Disadvantages of sexual reproduction

Fusion is difficult if two individuals are isolated.

Some variations may have undesirable qualities.

Population growth is slow

4.0 ENVIRONMENT

4.1 ECOSYSTEMS

What is an Ecosystem?

The ecosystem is the structural and functional unit of ecology where the living organisms interact with each other and the surrounding environment. In other words, an ecosystem is a chain of interactions between organisms and their environment.

Types of Ecosystem

An ecosystem can be as small as an oasis in a desert, or as big as an ocean, spanning thousands of miles. There are two types of ecosystem:

- Terrestrial Ecosystem
- Aquatic Ecosystem

Terrestrial Ecosystems

Terrestrial ecosystems are exclusively land-based ecosystems. There are different types of terrestrial ecosystems distributed around various geological zones. They are as follows:

- Forest Ecosystems
- Grassland Ecosystems
- Tundra Ecosystems
- Desert Ecosystem

Forest Ecosystem

A forest ecosystem consists of several plants, animals and microorganisms that live in coordination with the abiotic factors of the environment. Forests help in maintaining the temperature of the earth and are the major carbon sink.

Grassland Ecosystem

In a grassland ecosystem, the vegetation is dominated by grasses and herbs. Temperate grasslands, savanna grasslands are some of the examples of grassland ecosystems.

Tundra Ecosystem

Tundra ecosystems are devoid of trees and are found in cold climates or where rainfall is scarce. These are covered with snow for most of the year. The ecosystem in the Arctic or mountain tops is tundra type.

Desert Ecosystem

Deserts are found throughout the world. These are regions with very little rainfall. The days are hot and the nights are cold.

Aquatic Ecosystem

Aquatic ecosystems are ecosystems present in a body of water. These can be further divided into two types, namely:

- Freshwater Ecosystem
- Marine Ecosystem

Freshwater Ecosystem

The freshwater ecosystem is an aquatic ecosystem that includes lakes, ponds, rivers, streams and wetlands. These have no salt content in contrast with the marine ecosystem.

Marine Ecosystem

The marine ecosystem includes seas and oceans. These have a more substantial salt content and greater biodiversity in comparison to the freshwater ecosystem.

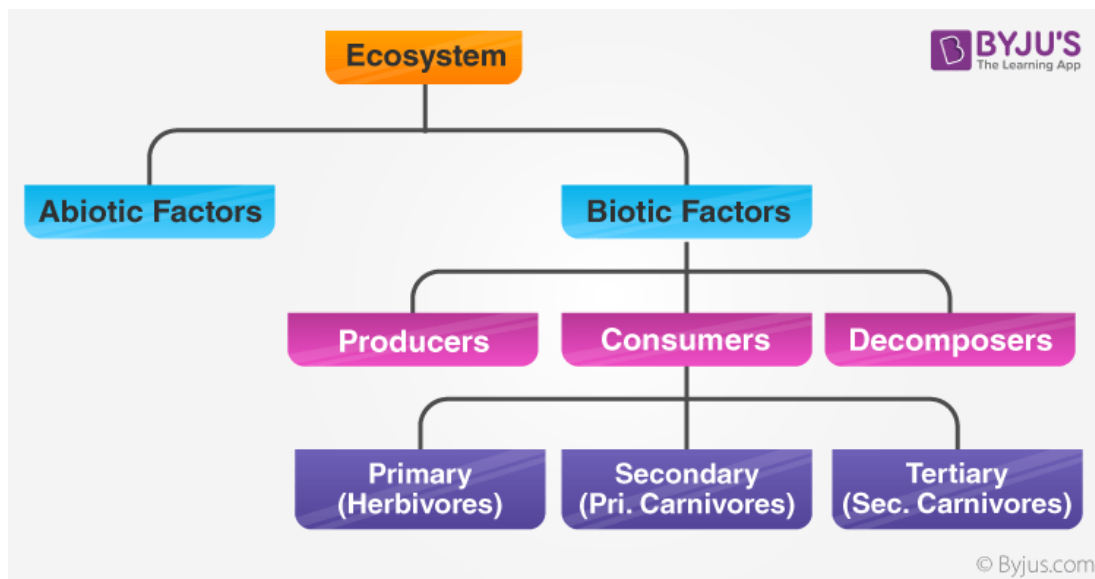
Structure of the Ecosystem

The structure of an ecosystem is characterised by the organisation of both biotic and abiotic components. This includes the distribution of energy in **our environment**. It also includes the climatic conditions prevailing in that particular environment.

The structure of an ecosystem can be split into two main components, namely:

- Biotic Components
- Abiotic Components

The biotic and abiotic components are interrelated in an ecosystem. It is an open system where the energy and components can flow throughout the boundaries.



Structure of Ecosystem highlighting the biotic and abiotic factors

Factors in an Ecosystem

1. Abiotic Factors (Environmental Factors)

Temperature

Is the hotness or coldness of an area or habitat.

It directly affects the distribution and productivity (yield) of populations and communities.

Most organisms are found in areas where temperature is moderate.

However, certain plants and animals have adaptations that enable them to live in areas where temperatures are in the extremes such as the hot deserts and the cold polar regions.

Temperatures not only influence distribution of organisms but also determine the activities of animals.

High temperature usually accelerates the rates of photosynthesis, transpiration, evaporation and the decomposition and recycling of organic matter in the ecosystem.

Light

Light is required by green plants for photosynthesis.

Light intensity, duration and quality affect organisms in one way or another.

Atmospheric Pressure

The force per unit area of atmospheric air that is exerted on organisms at different altitudes.

Growth of plants and activity of animals is affected by atmospheric pressure e.g., rate of transpiration in plants and breathing in animals.

Salinity

This is the salt content of soil or water.

Animals and plants living in saline conditions have special adaptations.

Humidity

This describes the amount of moisture (water vapour) in the air.

Humidity affects the rate of transpiration in plants and evaporation in animals.

pH

Is the measure of acidity or alkalinity of soil solution or water. pH

is very important to organisms living in water and soil.

Most prefer a neutral pH.

Wind:

Is moving air currents and it influences the dispersion of certain plants by effecting the dispersal of spores, seeds and fruits.

Air currents also modify the temperature and humidity of the surroundings.

Topography:

These are surface features of a place.

The topographical factors considered include altitudes, gradient (slope), depressions and hills. All these characteristics affect the distribution of organisms in an area e.g., the leeward and windward sides of a hill.

2. Biotic Components

Biotic components refer to all life in an ecosystem. Based on nutrition, biotic components can be categorised into autotrophs, heterotrophs and saprotrophs (or decomposers).

Producers include all autotrophs such as plants. They are called autotrophs as they can produce food through the process of photosynthesis. Consequently, all other organisms higher up on the food chain rely on producers for food.

Consumers or heterotrophs are organisms that depend on other organisms for food. Consumers are further classified into primary consumers, secondary consumers and tertiary consumers.

Primary consumers are always herbivores that they rely on producers for food.

Secondary consumers depend on primary consumers for energy. They can either be a carnivore or an omnivore.

Tertiary consumers are organisms that depend on secondary consumers for food. Tertiary consumers can also be an omnivore.

Quaternary consumers are present in some food chains. These organisms prey on tertiary consumers for energy. Furthermore, they are usually at the top of a food chain as they have no natural predators.

Decomposers include saprophytes such as fungi and bacteria. They directly thrive on the dead and decaying organic matter. Decomposers are essential for the ecosystem as they help in recycling nutrients to be reused by plants.

The biotic factors in an ecosystem include the following:

- competition,
- predation,
- symbiosis,
- parasitism,
- human activities.

Inter-relationships Between Organisms

The relationships between organisms in a given ecosystem is primarily a feeding one. Organisms in a particular habitat have different feeding levels referred to as trophic levels. There are two main trophic levels:

Producers

These organisms that occupy the first trophic level. They manufacture their own food hence are autotrophic.

Consumers

These are the organisms that feed on organic substances manufactured by green plants.

They occupy different trophic levels as follows:

Primary consumers:

These are herbivores and feed on green plants.

Secondary consumers:

These are carnivores and feed on flesh.

First order carnivores feed on herbivores while second order carnivores feed on other carnivores, i.e., tertiary consumers.

Omnivores:

These are animals that feed on both plant and animal material.

They can be primary, secondary or tertiary consumers.

Competition

This describes the situation where two or more organisms in the same habitat require or depend on the same resources.

Organisms in an ecosystem compete for resources like food, space, light, water and mineral nutrients.

Competition takes place when the environmental resource is not adequate for all.

Intraspecific competition.

This is competition between organisms of the same species.

For example, maize plants in a field compete for water and nutrients among themselves.

Interspecific competition

This refers to competition between organisms of different species, e.g., different species of predators can compete for water and prey among themselves.

Predation

It is a relationship whereby one animal (the predator) feeds on another (the prey).

Saprophytism

- Saprophytism is the mode of nutrition common in certain species of fungi and bacteria. Such organisms feed on dead organic material and release nutrients through the process of decomposition or decay.
- Saprophytes produce enzymes, which digest the substrates externally. The simpler substances are then absorbed.
- Saprophytes help in reducing the accumulation of dead bodies of plants and

animals. Harmful saprophytes cause rapid decay of foods such as fruits, vegetables, milk and meat. Others damage buildings by causing wood rot.

- Some fungi produce poisonous substances called aflatoxins.
- These substances are associated with cereal crops which are stored under warm, moist conditions. If the infected grain is eaten, it may cause serious illness, and death.

Parasitism

- This is an association between members of different species.
- The parasite lives on or in the body of another organism, the host.
- The parasite derives benefits such as food and shelter from the host but the host suffers harm as a result.

Symbiosis

This is an association in which organisms of different species derive mutual benefit from one another. Some symbiotic associations are loose and the two partners gain very little from each other. Other symbiotic associations are more intimate and the organisms show a high degree of interdependence.

Nitrogen Cycle

Is the interdependence of organisms on one another and the physical environment as nitrogen is traced from and back into the atmosphere

Although nitrogen is abundant in the atmosphere, most organisms are not able to utilise it directly. Some bacteria are capable of converting atmospheric nitrogen into forms which can be used by other living organisms.

These bacteria are referred to as nitrogen fixing bacteria.

Symbiotic nitrogen fixing bacteria live in the root nodules of leguminous plants such as beans and peas.

Non-symbiotic nitrogen fixing bacteria live in the soil. Nitrifying bacteria convert ammonia into nitrites and nitrates.

Denitrifying bacteria convert nitrates into atmospheric nitrogen.

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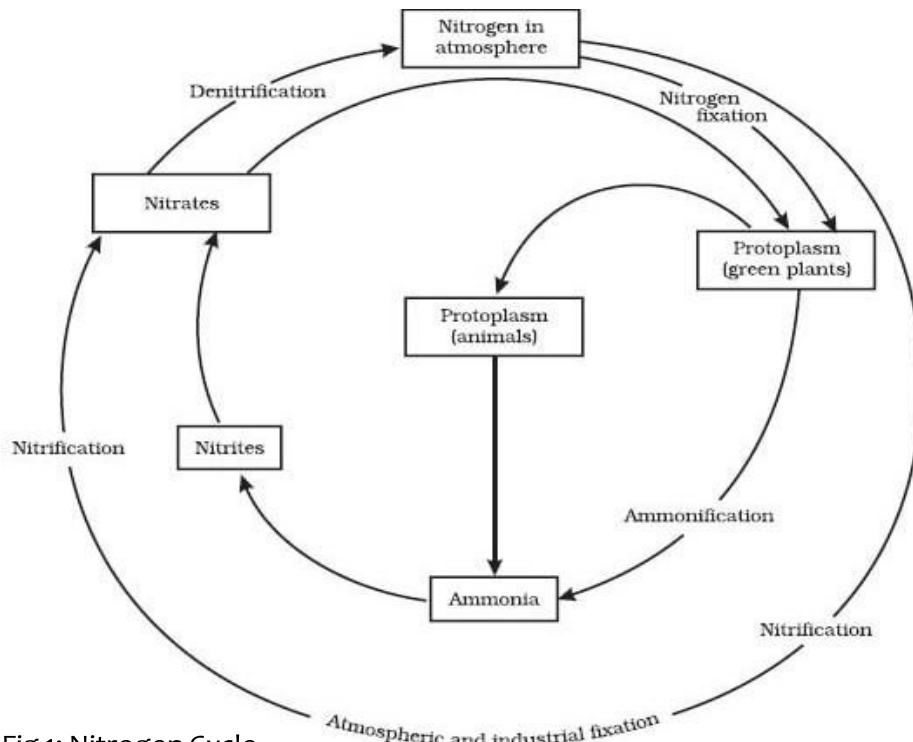


Fig 1: Nitrogen Cycle

Energy Flow in an Ecosystem

Most of the energy used in an ecosystem is derived from the sun. Solar energy is trapped by photosynthetic plants.

It flows through different trophic levels.

At each level energy is lost as heat to space and also through respiration.

Besides animals lose energy through excretion and defecation.

The amount of energy passed on as food from one trophic level to another decreases progressively.

The energy in the organisms is recycled back to plants through the various nutrient or material cycles.

Food Chains

A food chain is a linear relationship between producers and consumers.

It represents the transfer of food energy from green plants through repeated stages of eating and being eaten.

Types of Food Chains

Grazing food chain - starts with green plants.

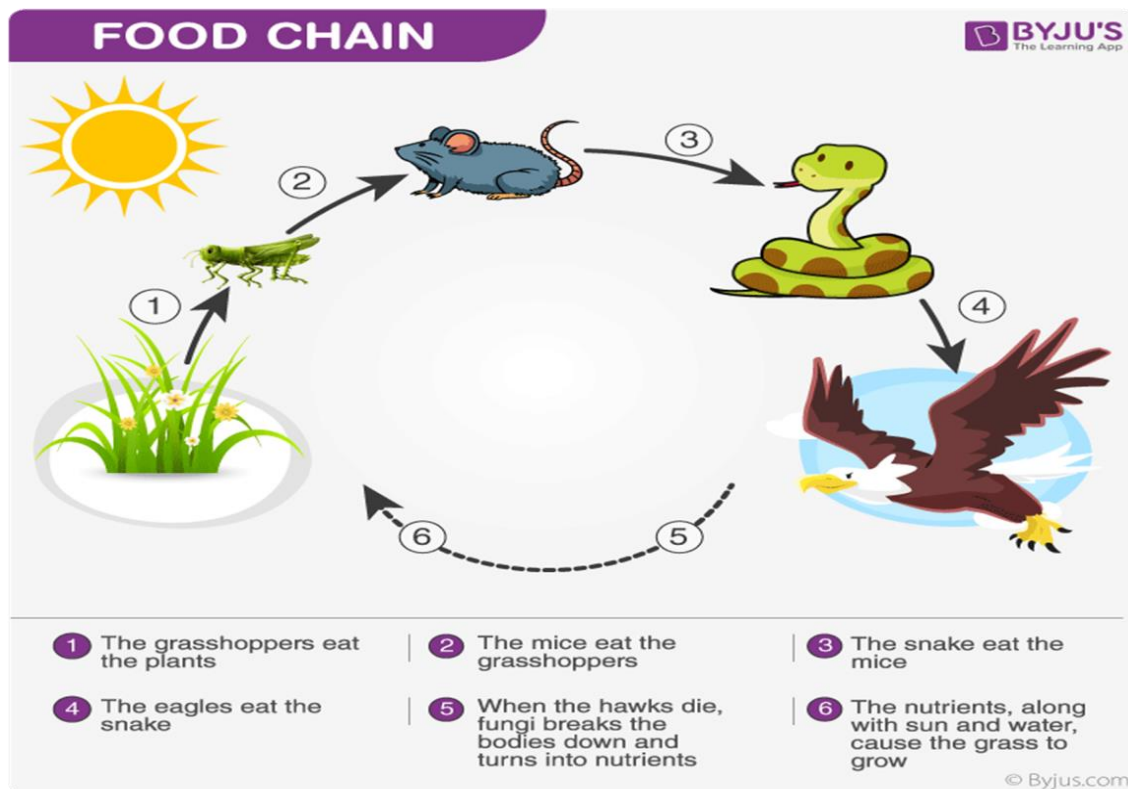
Detritus food chain - starts with dead organic material (debris or detritus).

Detritivores:

Detritivores feed on organic wastes and dead matter derived from the grazing food chain.

Many different types of organisms feed on detritus.

They include fungi, protozoa, insects, mites, annelids and nematodes.



Other examples of Food Chains

Green plants → aphids → lady-bird beetle

Green plants → antelope → lion

Algae → Tilapia → kingfisher

Plant debris → bacteria → eprotozoa → mosquito larva

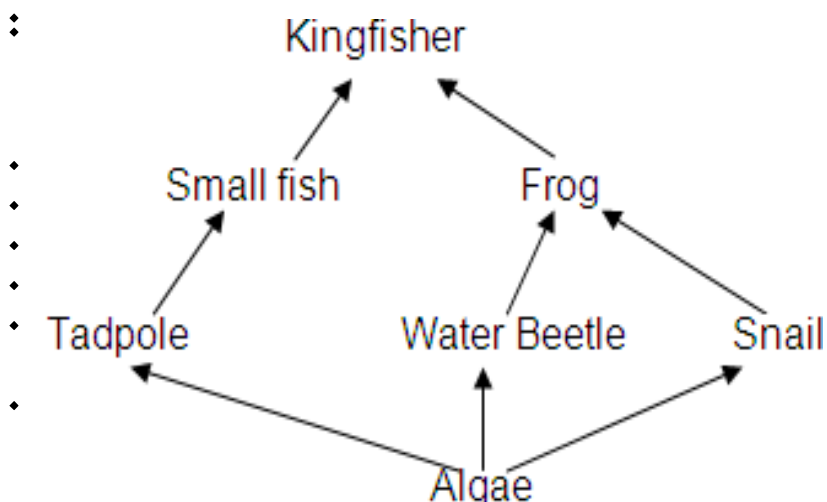
Phytoplankton → Zooplankton → Tilapia → Nile perch →

Human

Food Web

In a natural community, several food chains are interlinked to form a food web.

Several herbivores may feed on one plant.



Similarly, a given herbivore may feed on different plants and may in turn be eaten by different carnivores.

Fig 2: Food Web

Decomposers

These are mainly bacteria and fungi.

These organisms feed on dead organic matter thereby causing decomposition and decay and releasing nutrients for plants.

They form a link between the biotic and the abiotic components.

Functions of Ecosystem

The functions of the ecosystem are as follows:

- It regulates the essential ecological processes, supports life systems and renders stability.
- It is also responsible for the cycling of nutrients between biotic and abiotic components.
- It maintains a balance among the various trophic levels in the ecosystem.
- It cycles the minerals through the biosphere.
- The abiotic components help in the synthesis of organic components that involves the exchange of energy.

So the functional units of an ecosystem or functional components that work together in an ecosystem are:

Productivity – It refers to the rate of biomass production.

Energy flow – It is the sequential process through which energy flows from one trophic level to another. The energy captured from the sun flows from producers to consumers and then to decomposers and finally back to the environment.

Decomposition – It is the process of breakdown of dead organic material. The top-soil is the major site for decomposition.

Nutrient cycling – In an ecosystem nutrients are consumed and recycled back in various forms for the utilisation by various organisms.

Habitat and Adaptation

Habitat

Every organism has a unique [ecosystem](#) for its habitat. This ecosystem is its natural habitat. This is where the organism meets its basic need for its survival: food, water, shelter from the weather and place to breed its younglings.

The geographical feature and [environment](#) conditions on earth differ from one place to another. Some areas of the earth are mountains while others are covered by water. We also find snow, desert, forest, grassland, pond, rivers and lakes. There are many organisms that live in extreme environmental conditions.

Different Types of Habitat

1. Forest Habitat

Forest is a large area covered with plants. Forest covers about one-third of our planet. Most of the different types of plants and animals have their forest as their habitat. Forest broadly divides into three types:

Tropical forest or Tropical rain forest

These are found between the equator and two tropics, that is, Tropic of Cancer and Tropic of Capricorn. The Temperature of these forests ranges from 20 to 34 degree Celsius. These regions receive heavy rainfall throughout the year with annual rainfall of more than 200cm. These are spread in [India](#), Malaysia and other countries of South East Asia and South America.

Plants: Orchid, vine, moss, and fern.

Animals: Bat, gorilla, monkey, sloth, macaw, and variety of insects.



Temperate forest

Temperate [forests](#) are found in eastern North America, northeast Asia and western and central Europe. The temperature of these regions from ranges -30 to 30 Degree Celsius. The annual rainfall is about 150 cm. Temperate forests have well-defined winter and summer seasons.

Plants: Maple, oak, and elm.

Animals: Fox bald eagle, mountain lion, bobcat, and black bear.

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Boreal or Taiga forests

These forests are found in Canada, Russia, Scandinavia, China, Mongolia, and north Japan. Here, temperature dips from as low as -50degree Celsius to as high as 30 degree Celsius.

Plants: Evergreen trees such as pine, fir, deer, etc.

Animals: Wolf, lynx, fox, deer, wood packer, bat, and chipmunk.



2. Aquatic Habitat

Habitat in water is **aquatic habitats**. Such places include all the water bodies like rivers, lakes, seas, oceans. The organisms that live in water are **aquatic animals**. It is mainly of three kinds: freshwater, marine, and coastal.

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Freshwater habitat: Rivers, lakes, ponds, and streams are examples of freshwater habitat. Fish, frog, duck, lotus, and water lily are found in fresh [water](#),

Marine water habitat: Oceans and seas form the largest habitat on the planet. Some commonly found animals are whale, dolphins, sharks, octopus, starfish, jellyfish, seahorse, etc, and plants, seaweeds are the most common.

Coastal habitat: Coastal habitat refers to the region where the land meets the sea.

Estuaries: These places where the saline water of sea or ocean mixes with the fresh water of rivers. It is the home of marine animals like crab, oyster, worms, and waterfowl. Mangroves seaweeds and marsh grasses are plants found here.

Coral reefs: These are the rock-like structure made from calcium carbonate by corals. They are also called ‘rain-forest of the sea’. Some are commonly found animals are corals, starfish, sea anemone, octopus, etc.

3. Grassland Habitat

Grassland is regions dominated by grasses. There are too many trees and shrubs here. Temperature ranges between -20 degree and -30 degree. The annual rainfall varies between 50 cm and 90 cm. Some animals commonly in this habitat are giraffe, deer, zebra, lion, elephant, etc.



4. Dessert Habitat

Deserts are areas that receive very little rainfall less than 25 cm annually. In hot [desert](#), daytime temperature in summer can reach 45 degree Celsius. Organisms such as cactus, camel, gila monster and kangaroo rat are found here.



5. Mountainous and Polar Habitat

Polar and mountainous regions are the coldest of all the habitats found on earth. These include the Arctic and the Antarctic regions. In the Antarctic, the lowest temperature recorded was -88 Degree Celsius. Animals such as polar bear, penguin, seal and walrus are found in Polar Regions.



Adaptation

The [soil](#) and climate conditions on land are different in different parts of the earth. It is necessary for organisms to adjust to the conditions and environment they live in. The special characteristics possessed by plants and animals that enable them to successfully survive in a particular environment is the **adaptation**. Organism adapt to their habitat by the following means changes in:

Body

Behavior

Location

Adaptation In Tropical Forest

Plants: The following adaptation is shown by rainforest plants:

Leaves of tropical rainforest trees have specialized tips.

Due to the dense vegetation of rainforest, very little light is able to reach the forest floor. Plants growing in lower levels have big leaves to absorb as much sunlight as possible.

Animals: There is a huge variety of animals in rain forests.

- Many animals have adapted by learning to eat a particular food, which is eaten by no other animals.

Adaptation for Temperate Forests

Plants: The following are the adaptation shown by plants in temperate forests:

Most trees are deciduous. They have thin, broad leaves that allow them to easily capture [sunlight](#), which is required for making food.

Trees shed their leaves in winter because leaves are not able to bear the freezing winter.

Animals: Animals of temperate forests show the following adaptations:

- Many animals either hibernate or migrate to warmer regions during winter.
- Some animals gather food during summer and store it for eating during winter when food is scarce.



Adaptation for Boreal Forests

Plants: The following are the adaptations shown by plants in boreal forests:

Trees have a conical shape that allows the snow to slide off easily.

Trees have needle-like [leaves](#). This kind of structure protects the leaves from damage.

Animals: Animals in boreal forest show the following adaptations:

- Many animals migrate to warmer regions during winter. Some animals hibernate during winter months.
- Some animals have a thick layer of fur or feather to protect them from cold.

Adaptation for Grasslands

Plants: The following are the adaptations shown by the plants in grasslands:

- Grassland plants usually have flexible stems that bend instead of breaking when the wind is strong.
- Plants have strong roots that prevent winds from uprooting them.
- Plants have narrow or tiny leaves to reduce water loss.
- Some plants have roots that extend deep into the soil to absorb as much water as possible.

Animals: Animals in grassland show the following adaptations:

- Most grassland animals are able to run very fast. This ability also protects them from grassland fire.
- Many grassland animals have skin shades of brown that makes them hard to spot among the dry, brown grass.

Adaptation for Deserts

Plants: The following are the adaptations shown by the plants in grasslands:

- The leaves are modified as spines to minimize water loss.
- The stem is green, to make food for the plant.
- The stem is swollen and fleshy to store water.
- Cactus has a thick, waxy coating that prevents water loss and helps it to retain water.



Animals: Animals in desert show the following adaptations.

- Desert animals have thick skin to prevent the loss of water from the body.
- Most of the desert animals have the capacity to store water and food. For example, a camel can tolerate extremely hot temperature due to the stored water in its body, which helps in cooling.
- Most of the small desert animals live in burrows to save themselves from fluctuation in temperature.
- Reptiles are well-suited to the desert climates. They get most of the water through their food and lose hardly any moisture from their skin.

Adaptations in Mountain Regions

Plants: Plants in these regions have adapted themselves in the following ways:

- Mountain plants grow close to the ground to avoid being uprooted by strong winds.
- These plants also produce smaller leaves to prevent water loss.
- Some plants are also able to grow under a layer of snow.

Animals: Animals in the mountain and Polar Regions show following adaptations:

- Some mountain animals hibernate or migrate to warmer areas during colder months.
- Small ears and tail minimize heat loss from the body.
- Padded feet help the polar bear to walk on the snow.
- Thick fur and a layer of fat under the skin protect the polar bear from cold.

Adaptations for Aquatic Habitats

Plants: Aquatic plants show following adaptations:

- These plants have long, narrow stems. This prevents the plants from being carried away with the water current.
- Stems have air chambers that allow the aquatic plants to float in water leaves of plants such as lotus and water lily have a waxy coating that prevents them from rotting.
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Water Lilies



Cattails

Animals: aquatic animals show a variety of adaptations to survive in water:

- Ducks have webbed feet that help them in swimming.
- They also have hollow bones that help them to stay afloat.
- Gills are special organs that help fish to breathe underwater.
- They have streamlined body which allows them to swim fast by reducing resistance due to flowing water.
- Dolphins and whale have **blowholes** at the upper parts of their heads. They come to the water surface and breathe in air through the blowholes from time to time.



Adaptation and Acclimatization

Adaptation refers to change in an organism over a long period of time. There are certain changes that can occur in an organism over a short period of time, which help the organisms adjust to the changes in its surrounding. This is Acclimatization.

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Adaptations of Plants to Various Habitats

Organisms have developed structural features that enable them to live successfully in their particular habitats.

Plants found beneath the canopies of trees are adapted to low light intensities by having broad leaves.

Xerophytes

These are plants that grow in dry habitats, i.e., in deserts and semi-deserts.

They have adaptations to reduce the rate of transpiration in order to save on water consumption.

Others have water storage structures.

Adaptations include:

- Reduction of leaf surface area by having needle-like leaves, rolling up of leaves and shedding of leaves during drought to reduce water loss or transpiration.
- Thick cuticle; epidermis consisting of several layers of cells; leaves covered with wax or resin to reduce evaporation.
- Sunken stomata, creating spaces with humid still air to reduce water loss. Few, small stomata, on lower epidermis to reduce water loss.
- Stomata open at night (reversed stomatal rhythm) to reduce water loss. Deep and extensive root systems for absorption of water.

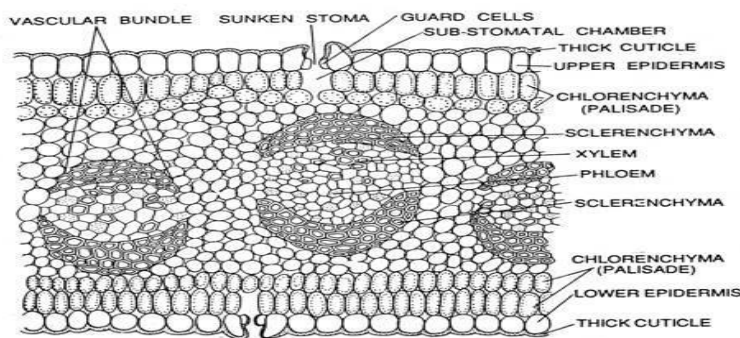


Fig. 41.32. Anatomy of xerophytic leaf. V.S. of leaf of *Agave* (monocot) showing detailed structure.

Development of flattened shoots and succulent tissue for water storage e.g. *Opuntia*.

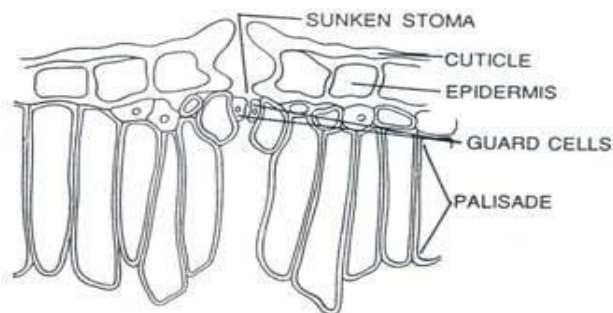
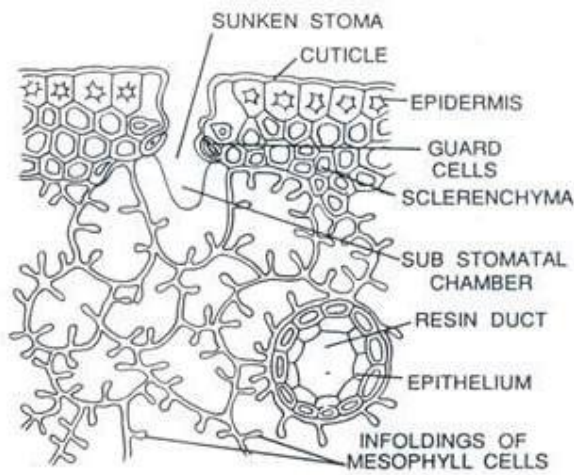


Fig. 41.34. Structure of xerophytic leaf. T.S. outer layers of cells of *Hakea* leaf.



Structure of xerophytic leaf (microphyllous type). C.S. of a marginal part of leaf of *Pinus nigra*.

Mesophytes

- These are the ordinary land plants which grow in well-watered habitats. They have no special adaptations.
- Stomata are found on both upper and lower leaf surfaces for efficient gaseous exchange and transpiration.
- However, those found in constantly wet places e.g. tropical rain forests, have features that increase transpiration.
- These plants are called hygrophytes.
- The leaves are broad to increase surface areas for transpiration and thin to ensure short distance for carbon (IV) oxide to reach photosynthetic cells and for light penetration.
- The stomata are raised above the epidermis to increase the rate of transpiration.
- They have glandular hairs or hydathode that expel water into the saturated atmosphere. This phenomenon is called guttation.

Hydrophytes (Water Plants)



Water Lilies



Cattails

Water plants are either submerged, emergent or floating.

Submerged Plants

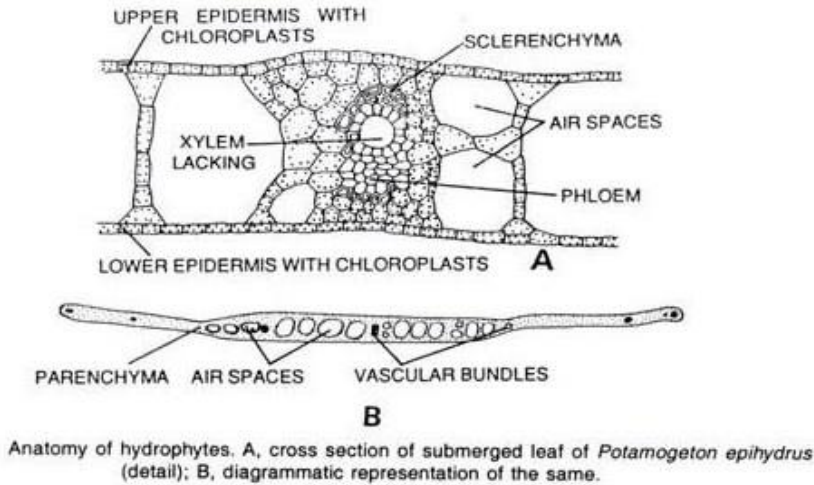
- The leaves have an epidermis with very thin walls and a delicate

cuticle. They have no stomata.

- Water is excreted from special glands and pores at the tips.
- Presence of large air spaces and canals (aerenchyma) for gaseous exchange and buoyancy. Some plants have filamentous leaves in order to increase the surface area for absorption of light, gases and mineral salts.
- Some plants are rootless, hence support provided by water.

Mineral salts and water absorbed by all plant surfaces.

In some plants, the stem and leaves are covered with a waxy substance to reduce absorption of water. e.g. *Ceratophyllum* and *Elodea* sp.



Floating Plants

- Their structure is similar to that of mesophytes.
- The leaves are broad to increase the surface area for water loss.
- They have more stomata on the upper surface than on the lower surface to increase rate of water loss.

Examples are *Pistia* sp. (water lettuce), *Salvinia* and *Nymphaea*.

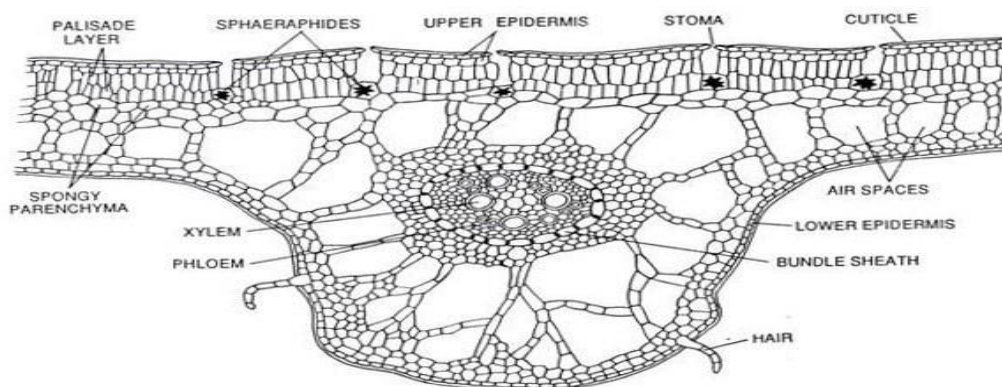


Fig. 41.25. Hydrophytes. T.S. floating leaf of *Trapa bispinosa*, showing big air spaces; the stomata confined to upper epidermis only.

Halophytes (Salt Plants)

- These are plants that grow in salt marshes and on coastlines.
- They have root cells that concentrate salts and enable them to take in water by osmosis. They have salt glands which excrete salts.
- Fruits have large aerenchymatous tissues for air storage that makes them float. Some have shiny leaves to reduce water loss.
- The mangrove plants have roots that spread horizontally, and send some branches

into the air. These aerial roots are known as breathing roots or pneumatophores.

- They have lenticel-like openings called pneumatophores through which gaseous exchange takes place.

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4.2 ENVIRONMENTAL ACTIVITIES

Pollution

This is the introduction of foreign material, poisonous compounds and excess nutrients or energy to the environment in harmful proportions.

Any such substance is called a pollutant.

Effects and Control of Causes of Pollutants in Air, Water and Soil

Industrialisation and urbanisation are the main causes of pollution.

As human beings exploit natural resources the delicate balance in the biosphere gets disturbed. The disturbance leads to the creation of conditions that are un-favourable to humans and other organisms.

Sources of Pollutants

- Motor vehicles release carbon (II) oxide, sulphur (IV) oxide, and nitrogen oxides and hydrocarbons. Agricultural chemicals, fertilisers and pesticides.
- Factories, manufacturing and metal processing industries.
- They release toxic substances and gases as well as synthetic compounds that are bio-undegradable.
- They release solid particles or droplets of poisonous substances e.g. arsenic, beryllium, lead and cadmium.
- Radioactive waste: Leakages from nuclear power stations and testing sites release radioactive elements like strontium-90 which can eventually reach man through the food chain.
- Domestic waste and sewage are released raw into water bodies.
- Oil spills from accidents in the seas and leakage of oil tankers as well as from offshore drilling and storage and processing.

Water Pollution

In most cases, toxic chemical wastes from industries are discharged into water.

Toxic chemicals such as mercury compounds may be ingested by organisms.

Insecticides like DDT, and weedkillers eventually get into the water and contaminate it.

Oil and detergents also pollute water.

Excess nitrates and phosphates from sewage and fertilisers cause overgrowth of algae and bacteria in water.

This is called eutrophication.

As a result there is insufficient oxygen which causes the deaths of animals in the water.

Air Pollution

Smoke from industries and motor vehicles contains poisonous chemicals like carbon (II) oxide, carbon (IV) oxide, sulphur (IV) oxide and oxides of nitrogen.

When sulphur (IV) oxide and oxides of nitrogen dissolve in rain, they fall as acid rain.

Accumulation of carbon (IV) oxide in the atmosphere causes the infrared light to be confined within the atmosphere, the earth's temperature rises.

This is called the greenhouse effect.

Carbon particles in smoke coat the leaves of plants and hinder gaseous exchange and photosynthesis.

The particles also form smog in the air.

Lead compounds are from vehicle exhaust pipes.

All these have negative effects on man and the environment.

Soil/Land pollution

Plastics and other man-made materials are biologically non-degradable i.e they are not acted upon by micro-organisms.

Scrap metal and slag from mines also pollute land.

Failure to rehabilitate mines and quarries also pollute land.

Effects of Pollutants to Humans and Other Organisms

Chemical pollutants e.g. nitrogen oxides, fluorides, mercury and lead cause physiological and metabolic disorders to humans and domestic animals.

Some hydrocarbons as well as radioactive pollutants act as mutagens (cause mutations) and carcinogens induce cancer.

Radioactive pollutants like strontium, caesium and lithium are absorbed into body surface and cause harm to bone marrow and the thyroid gland.

Communicable diseases like cholera are spread through water polluted with sewage. Thermal pollution results in death of some fish due to decreased oxygen in the water.

Oil spills disrupt normal functioning of coastal ecosystems.

Birds that eat fish die due to inability to fly as feathers get covered by oil.

Molluscs and crustaceans on rocky shores also die.

Control of Air Pollution

Use of lead-free petrol and low sulphur diesel in vehicles.

Use of smokeless fuels e.g electricity or solar.

Filtration of waste gases to remove harmful gases. Liquid dissolution of waste gases.

In Kenya, factories are subjected to thorough audits to ensure that they do not pollute the

environment.

Factories should be erected far away from residential areas.

Reduce volume or intensity of sound.

Use of ear muffs.

Vehicle exhaust systems should be fitted with catalytic oxidisers.

Regular servicing of vehicles to ensure complete combustion of fuel.

Control of Water Pollution

Treatment of sewage.

Treatment of industrial waste before discharge into water.

Use of controlled amounts of agrochemicals.

Organic farming and biological control.

Avoid spillage of oils and other chemicals into water.

Good water management.

Stiff penalties for oil spillage.

Use of Pseudomonas bacteria that naturally feed on oil and break it up.

Control of Soil Pollution

Addition of lime to farms to counteract the effect of agrochemicals.

Recycling of solid waste.

Compacting and incineration of solid waste. Use of biodegradable materials and chemicals. Good soil management to avoid soil erosion.

Human Diseases

The term disease denotes any condition or disorder that disrupts the steady state of well being of the body.

Health is a state of physical, mental and emotional well being in the internal environment of the body.

Some of the causes of diseases are due to entry of pathogens and parasites.

Pathogens include bacteria, viruses, protozoa and fungi.

Parasites are organisms which live on or in the body of another organisms.

Vectors are animals that carry the pathogen from one person to another.

Most are ectoparasites that transmit the disease as they feed.

Bacterial Diseases

Cholera

Causative agent a bacterium Vibrio cholerae.

Transmission - It is spread through water and food contaminated by human faeces containing the bacteria.

The bacteria produce a powerful toxin, enterotoxin, that causes inflammation of the wall of the intestine leading to:

Severe diarrhoea that leads to excessive water loss from body. Abdominal pain

Vomiting

Dehydration which may lead to death.

Prevention and Control

Adequate sanitation such as water purification sewage treatment and proper disposal of human faeces.

Public and personal hygiene e.g washing hands before meals and washing fruits and vegetables, boiling drinking water.

Vaccination

Carriers should be identified, isolated and treated during outbreaks.

Treatment

Use of appropriate antibiotics.

Correcting fluid loss by injecting fluids or by administration of oral rehydration solutions.

Typhoid

Causative agent.

The disease is caused by *Salmonella typhi*.

Transmission is through contaminated water and food.

It is also transmitted by certain foods, e.g. oysters, mussels and shell fish.

Symptoms

- Fever
 - ↳ Muscle pains
 - ↳ Headache
- Spots on the trunk of the body
- Diarrhoea
- In severe cases mental confusion may result and death.

Prevention

- Boil drinking water.
- Proper sewage treatment.
- Proper disposal of faeces, if not flushed use deep pit latrines. Observe personal hygiene e.g. washing hands before meals. Washing fruits and vegetables.

Treatment

Use of appropriate antibiotics.

Protozoa Diseases

Malaria

Malaria is caused by the protozoan plasmodium.

The most common species of plasmodium are *P. falciparum*, *P. vivax*, *P. malariae* and *P. ovale* with varying degree of severity.

Transmission

Is by female anopheles mosquito as it gets a blood meal. Symptoms

Headache, sweating, shivering, high temperature (40-41 °C) chills and joint pains.

The abdomen becomes tender due to destruction of red blood cells by the parasites.

Prevention

Destroy breeding grounds for mosquitoes by clearing bushes and draining stagnant water.

Kill mosquito larvae by spraying water surfaces with oil.

Use insecticides to kill adult mosquitoes

Sleeping under a mosquito net.

Take preventive drugs.

Treatment

Use appropriate anti-malarial drugs.

Amoebic dysentery (Amoebiasis)

Cause

This disease is caused by *Entamoeba histolytica*.

The parasites live in the intestinal tract but may occasionally spread to the liver.

Transmission - They are transmitted through contaminated water and food especially salads.

Symptoms

Abdominal pain, nausea and diarrhoea.

The parasites cause ulceration of the intestinal tract, which results in diarrhoea.

Prevention and control

Proper disposal of human faeces. Boiling water before drinking.

Personal hygiene e.g. washing hands before meals.

Washing vegetables and steaming particularly salads and fruits before eating.

Treatment

Treatment of infected people with appropriate drugs.

Parasitic Diseases

Ascaris lumbricoides



Ascaris lumbricoides lives in the intestines of a man or pig, feeding on the digested food of the host.

The body of the worm is tapered at both ends. The female is longer than the male. Mode of transmission

The host eats food contaminated with the eggs, the embryo worms hatch out in the intestine. The embryo worms then bore into the blood vessels of the intestine.

They are carried in the bloodstream to the heart and then into the lungs.

As they travel through the bloodstream, they grow in size.

After sometime, the worms are coughed out from the air passages and into the oesophagus.

They are then swallowed, eventually finding their way into the intestines where they grow into mature worms.

Effects of Ascaris lumbricoides on the host

The parasites feed on the host's digested food. This results in malnutrition especially in children.

If the worms are too many, they may block the intestine and interfere with digestion.

The worms sometimes wander along the alimentary canal and may pass through the nose or mouth.

In this way, they interfere with breathing and may cause serious illness.

The larvae may cause severe internal bleeding as they penetrate the wall of the intestine.

Adaptive Characteristics

The female lays as many as 25 million eggs. This ensures the continuation of the species.

Eggs are covered by a protective cuticle that prevents them from dehydration.

The adult worms tolerate low oxygen concentration.

Have mouth parts for sucking food and other fluids in the intestines.

Has a thick cuticle or pellicle to protect it from digestive enzymes produced by the host.

Control and Prevention

Personal hygiene e.g. washing hands before eating.

Proper disposal of faeces.

Washing of fruits and vegetables.

Treatment

Deworm using appropriate drugs ant-helminthics.

Schistosoma

Schistosoma or bilharzia worm is a flat worm, parasitic on human beings and fresh water snails.

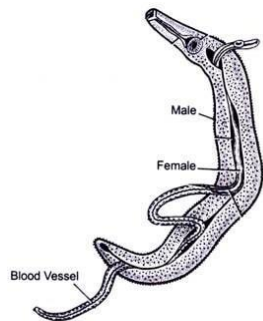


Fig. 9.18 Schistosoma male & female in pair

(Biomphalaria and Bulinus.) The snail act as intermediate host.

Mode of Transmission

- Schistosomiasis also known as a bilharziasis is caused by several species of the genus Schistosoma.
- *Schistosoma haematobium* infects the urinary system mainly the bladder
- *S. japonicum* and *S. mansoni* both infect the intestines.
- *Schistosoma haematobium* is common in East Africa where irrigation is practised and where slow moving fresh water streams harbour snails.
- It is spread through contamination of water by faeces and urine from infected persons.
- The embryo (miracidium) that hatch in water penetrates into snails of the species *Biomphalaria* and *Bulinus*.
- Inside the snail's body, the miracidium undergoes development and multiple fission to produce rediae.
- The rediae are released into the water and develop to form cercariae which infect humans through:
 - Drinking the water
 - Wading in water;
 - Bathing in snail-infested water.
- The cercaria burrows through the skin and enters blood vessel.

Effects on the host

- Inflammation of tissues where egg lodge. Ulceration where eggs calcify.
- Egg block small arteries in lungs leading to less aeration of blood. The body turns blue - a condition known as cyanosis.
- If eggs lodge in heart or brain, lesions formed can lead to death.
- Bleeding occurs as the worms burrow into blood vessels (faeces or urine has blood). Pain and difficulty in passing out urine.
- Nausea and vomiting.

When eggs lodge in liver ulceration results in liver cirrhosis. Death eventually occurs. **Adaptive Characteristics**

- The female has a thin body and fits into small blood vessels to lay eggs. Eggs are able to burrow out of blood vessel into intestine lumen.
- Many eggs are laid to ensure the survival of the parasite. Large numbers of cercariae are released by snail.
- The miracidia and cercariae larvae have glands that secrete lytic enzymes which soften the tissue to allow for penetration into host.
- The male has a gynecophoric canal that carries the female to ensure that eggs are fertilised before being shed.
- Has suckers for attachment.

Prevention and Control

- Drain all stagnant water
- Boil drinking water.
- Do not wade bare feet in water.
- Wear long rubber boots and gloves (for those who work in rice fields). Eliminate snails, by spraying with molluscides
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